CHAPTER 73
FUEL

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CHAPTER 73
FUEL

73-00-00 INTRODUCTION

The fuel system supplies clean fuel at the correct pressure and flow to the fuel nozzles in order to support combustion throughout the full operating range of the engine.

GENERAL

Three sub-systems comprise the fuel system:

- Engine Fuel Distribution
- Engine Fuel Control and
- Engine Fuel Indication.

They will be explained in detail in this chapter.

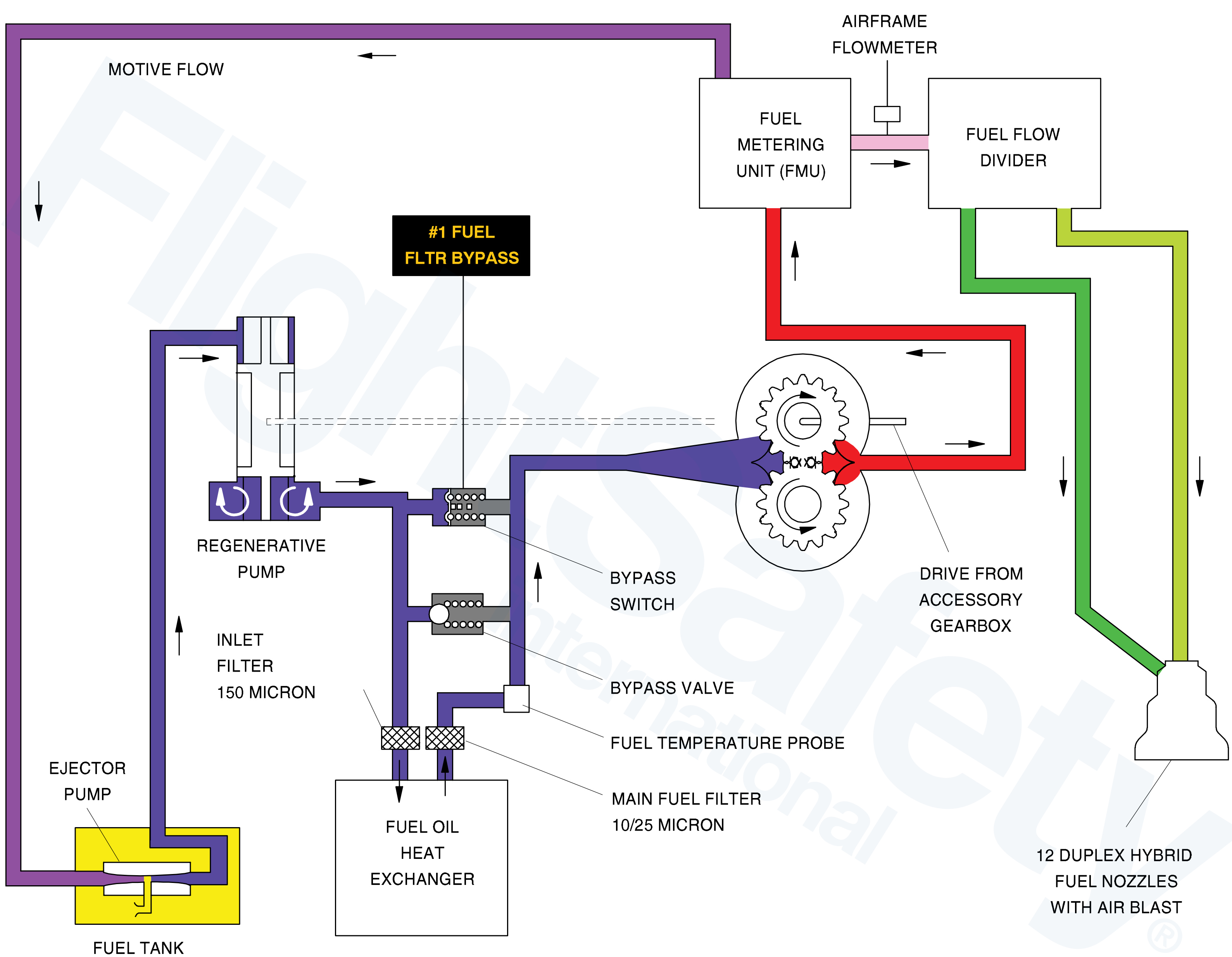

Figure 73-1. Engine Fuel Control Diagram

SYSTEM DESCRIPTION

Refer to Figure 73-1. Engine Fuel Control Diagram.

The fuel is received by the FMU at the regenerative fuel-pump inlet-port. The regenerative fuel pump supplies fuel to the fuel heater. The fuel is heated (if necessary), filtered and returned to the FMU for metering.

The FMU controls the fuel flow supplied to the engine based on demand, from the FADEC.

The FADEC calculates the required amount of fuel to supply based on various engine sensor inputs like N_H, N_L, N_P engine temperature and torque calculations.

The FADEC acts on the metering valve in the FMU to control the fuel flow. The metered fuel is then routed to the airframe-supplied flowmeter and to the flow divider valve through external tubes. Some of the metered fuel is also returned to the airframe fuel tanks to drive the main ejector pump.

The flow divider valve takes metered fuel and distributes the fuel to the primary and secondary fuel manifolds. During start, when the fuel pressure coming from the FMU is low, only the primary manifold is supplied with fuel.

As the fuel pressure increases, the secondary manifold is also supplied with fuel. The flow divider valve also recoups the remaining fuel in the manifolds on engine shutdown. This fuel is then used on the next engine start.

The fuel manifolds take the fuel from the flow divider valve and supply the twelve fuel nozzle adapters. All fuel nozzle adapters are duplex units and have both primary and secondary passages.

NOTES

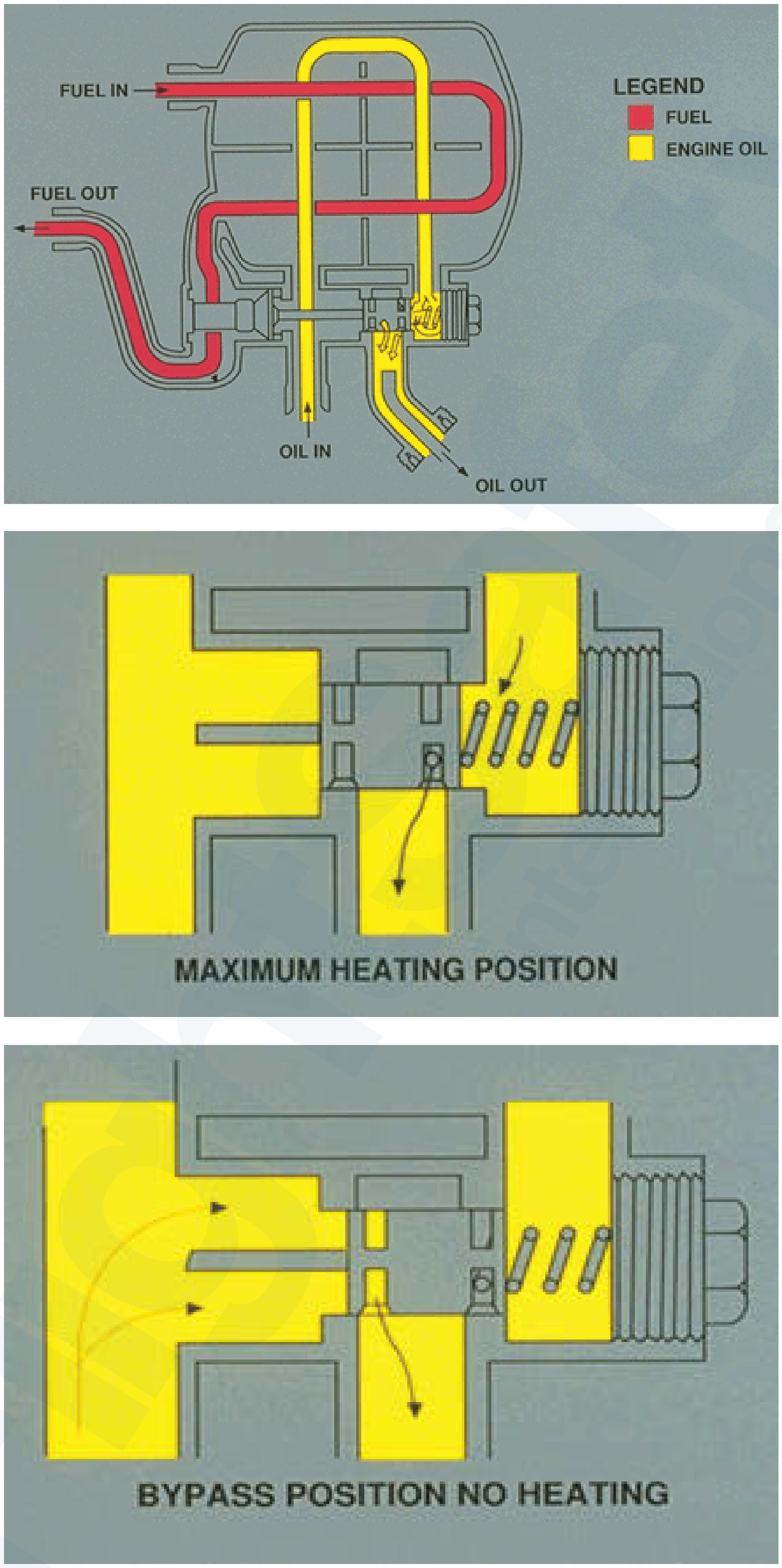

Figure 73-2. Oil to Fuel Heat Exchanger (Flow)

73-11-00 ENGINE FUEL DISTRIBUTION

GENERAL

The engine receives fuel from the airframe fuel system.

The fuel passes through the fuel heater and is pumped by an integrated (regenerative) fuel pump in the FMU.

The fuel is then metered by the FMU. The metered fuel goes to the flow divider valve which separates it into primary and secondary for delivery to the fuel nozzles through the fuel manifolds.

The fuel distribution system contains the components listed below:

- Oil to fuel heat exchanger
- Fuel heater oil pressure relief valve
- Fuel strainer
- Fuel heater thermal actuator valve
- Fuel nozzles and manifold
- Fuel nozzle adapter
- Flow divider valve and
- Fuel filter.

SYSTEM DESCRIPTION

Refer to Figure 73-2. Oil to Fuel Heat Exchanger (Flow).

Fuel for the engine passes through the oil to fuel heater to prevent ice forming in the fuel.

The fuel passes from the fuel heater through the fuel filter to remove any contaminant particles.

The fuel then passes to the flow divider where it is directed to the primary and secondary fuel manifolds.

The manifolds deliver the fuel to the nozzles, where it is atomized and sprayed into the combustor.

LEGEND
1. Transfer tube assembly
2. Transfer tube
3. Transfer tube
4. Fuel heater
5. Ground cable
6. Oil filter
7. FMU
8. Fuel temperature sensor

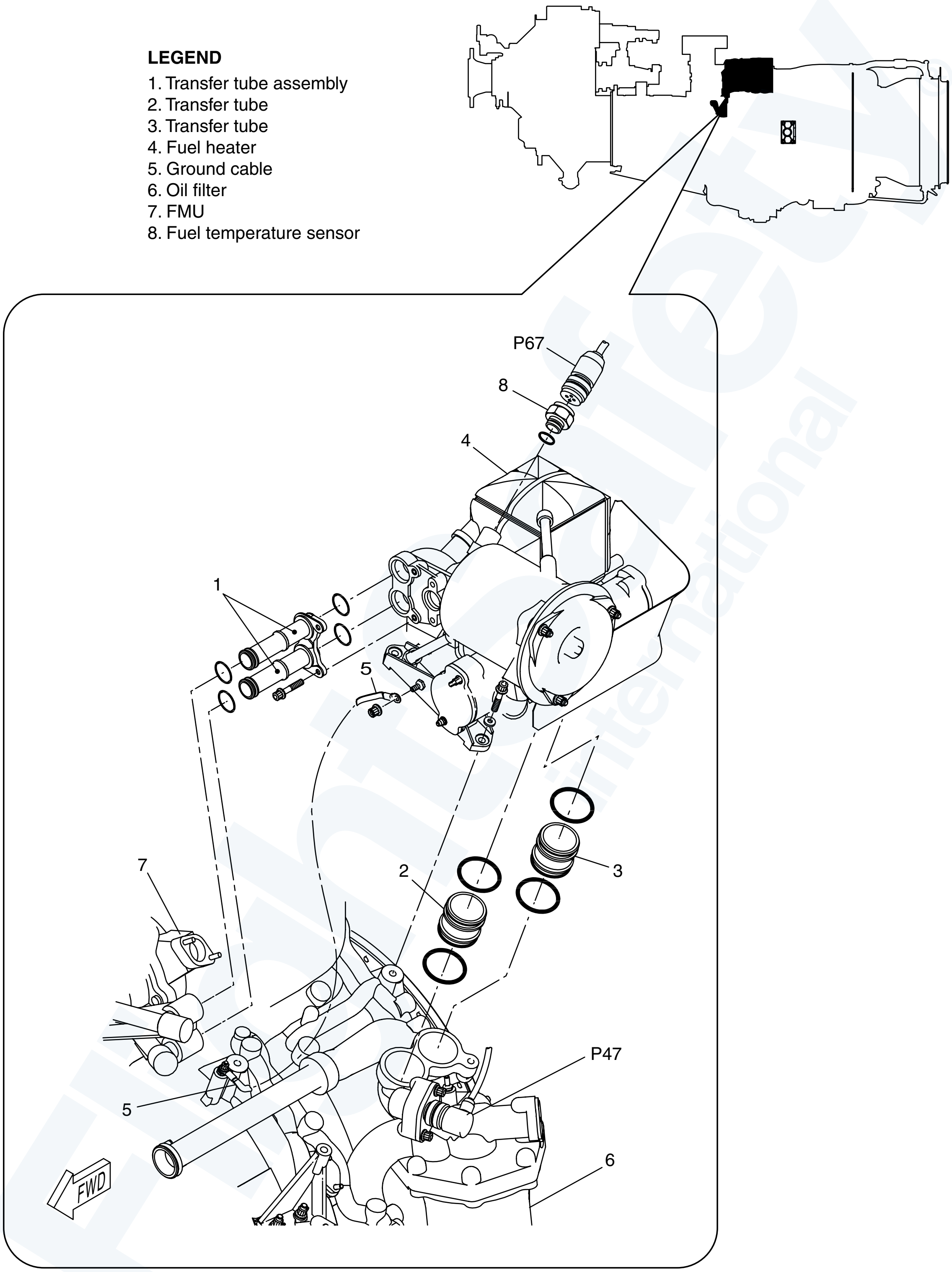

Figure 73-3. Oil to Fuel Heat Exchanger

COMPONENT DESCRIPTION

Oil to Fuel Heat Exchanger

Refer to:

- Figure 73-2. Oil to Fuel Heat Exchanger (Flow).
- Figure 73-3. Oil to Fuel Heat Exchanger.
- Figure 73-4. Oil to Fuel Heat Exchanger.

The oil to fuel heat exchanger, installed on the low pressure compressor case of the engine, it heats the fuel to prevent the formation of ice crystals.

The heat exchanger is an integral assembly. It is made of aluminum castings consisting of, a heater assembly, and a filter assembly. Both assemblies are welded together and reinforced with struts.

It is connected to the Fuel Metering Unit (FMU) through two transfer tubes on the forward face of the fuel heater. An additional two transfer tubes at the bottom of the heater connect it to the main oil-filter-housing.

The heat transfer matrix is comprised of an aluminum plate and fin Assembly that separates oil and fuel from each other in passages.

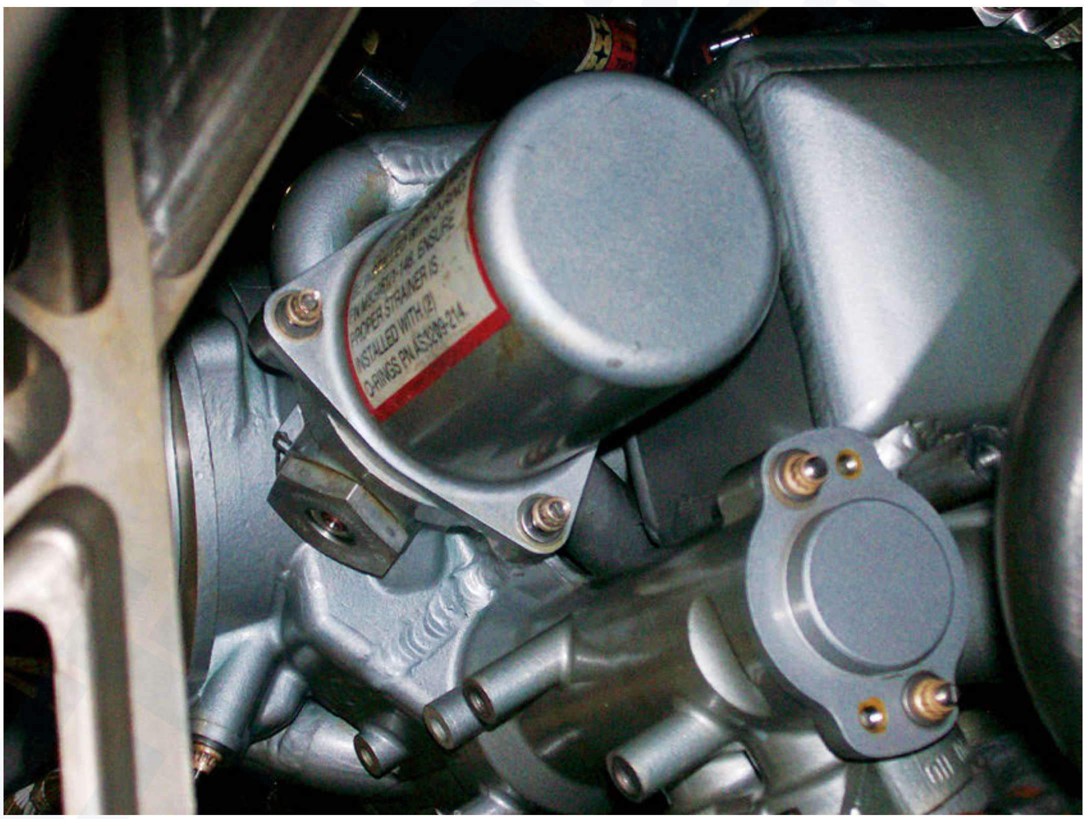

Figure 73-4. Oil to Fuel Heat Exchanger

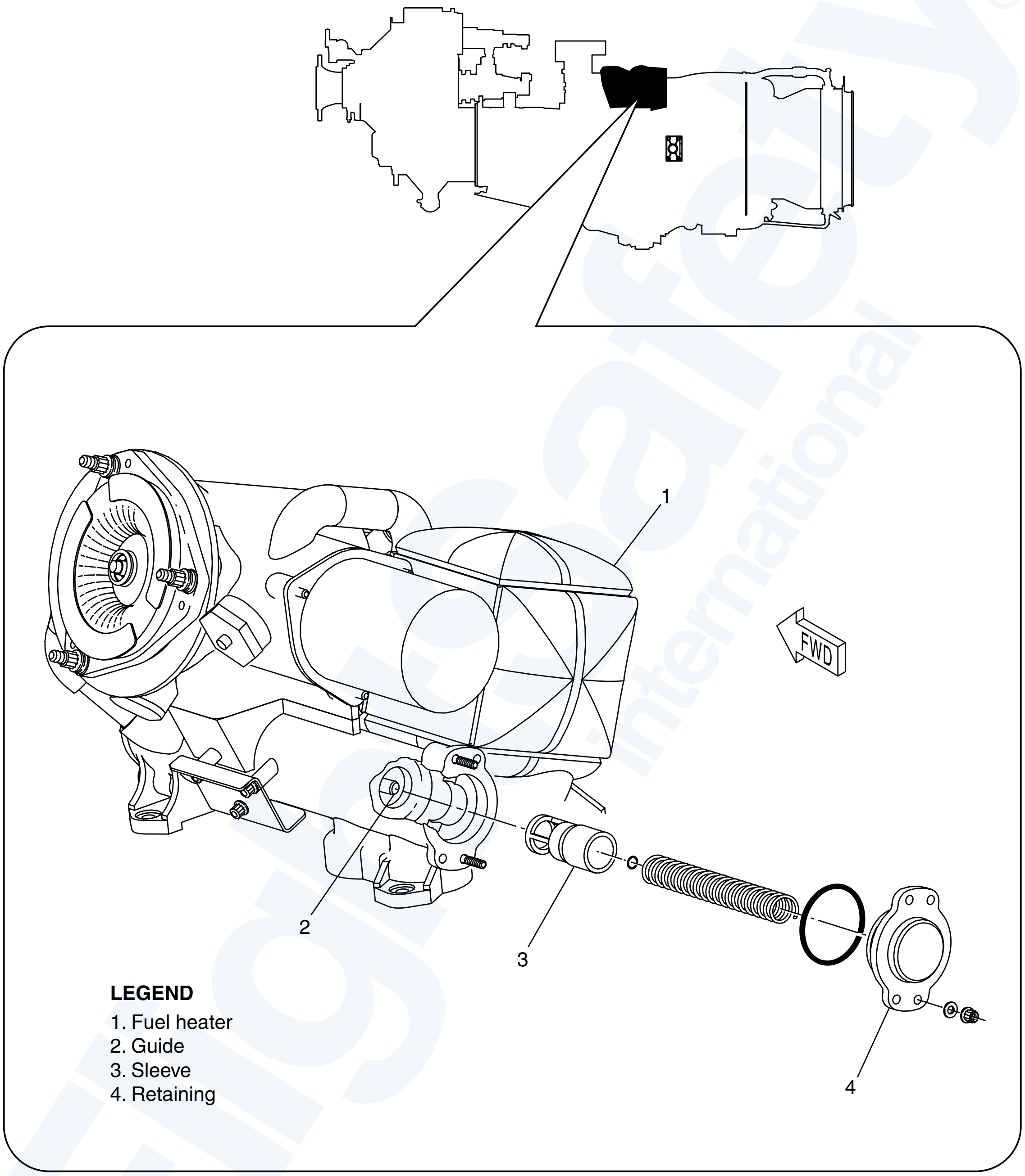

Figure 73-5. Fuel Heater Oil Pressure Relief Valve

Fuel Heater Oil Pressure Relief Valve

Refer to Figure 73-5. Fuel Heater Oil Pressure Relief Valve.

The Oil Pressure relief valve is a spring relief valve. It is in the fuel heater as depicted in 3.

The fuel heater incorporates an oil bypass valve on the oil side of the unit which is both pressure and temperature controlled. The oil bypass valve will bypass oil when the oil pressure reaches 28 ±3 psid (193 ±21 kPad). This protects the fuel heater in the event that oil passages are blocked in the fuel heater, or if the oil pressure becomes too high.

NOTES

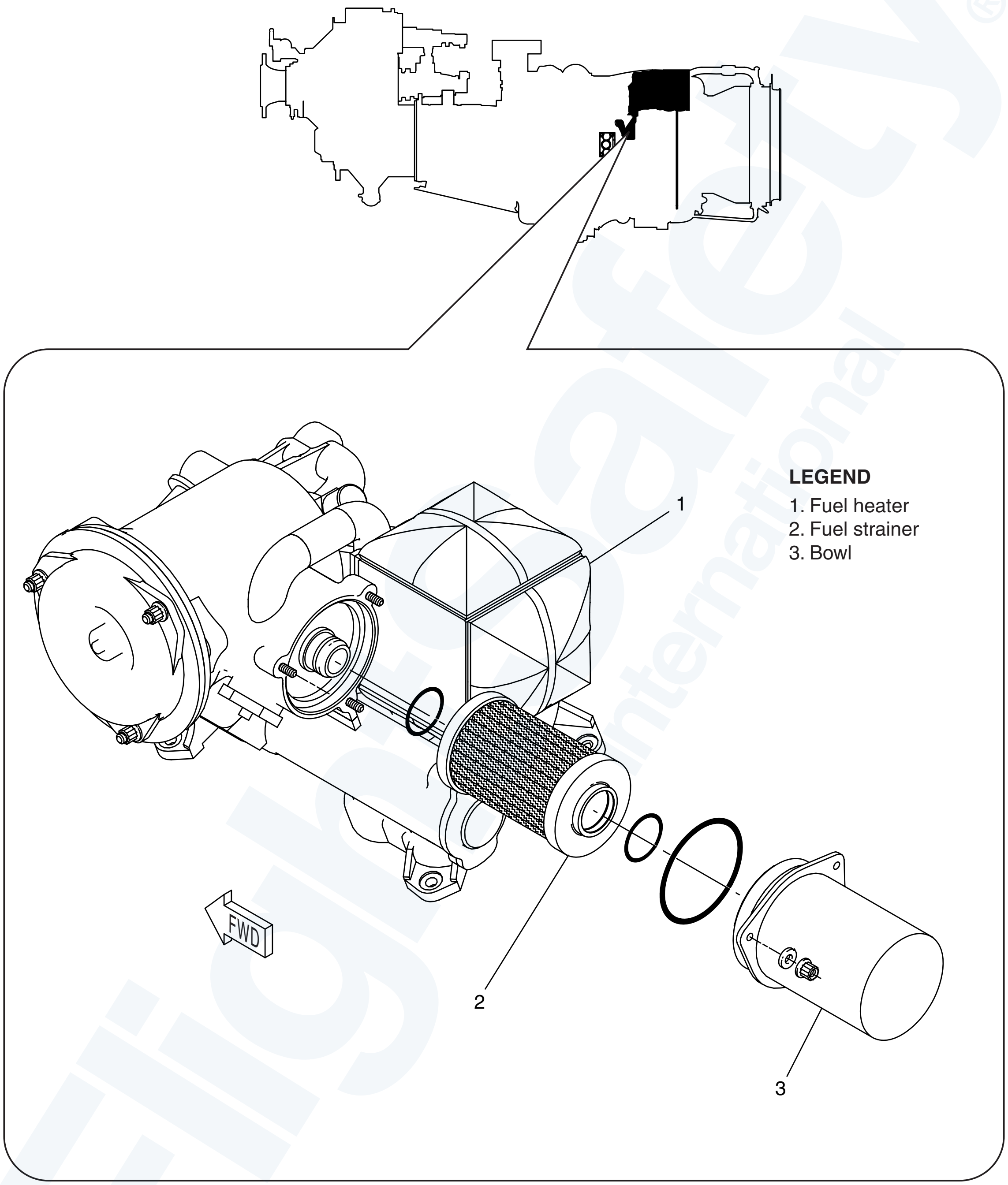

Figure 73-6. Fuel Strainer

Fuel Strainer

Refer to:

- Figure 73-6. Fuel Strainer.
- Figure 73-7. Fuel Strainer.

The fuel strainer is attached to the fuel heater. It is a 150 micron absolute strainer installed upstream of the heat transfer matrix.

The fuel strainer prevents contamination of the matrix while being resistant to freezing of any water contained in the fuel. This strainer is protected by a bypass valve equipped with a bypass indicator.

Figure 73-7. Fuel Strainer

LEGEND
1. Thermal actuator
2. Poppet assembly
3. Connector P41 (REF)
4. Fuel heater
5. Connector P67 (REF)
6. Connector P47 (REF)

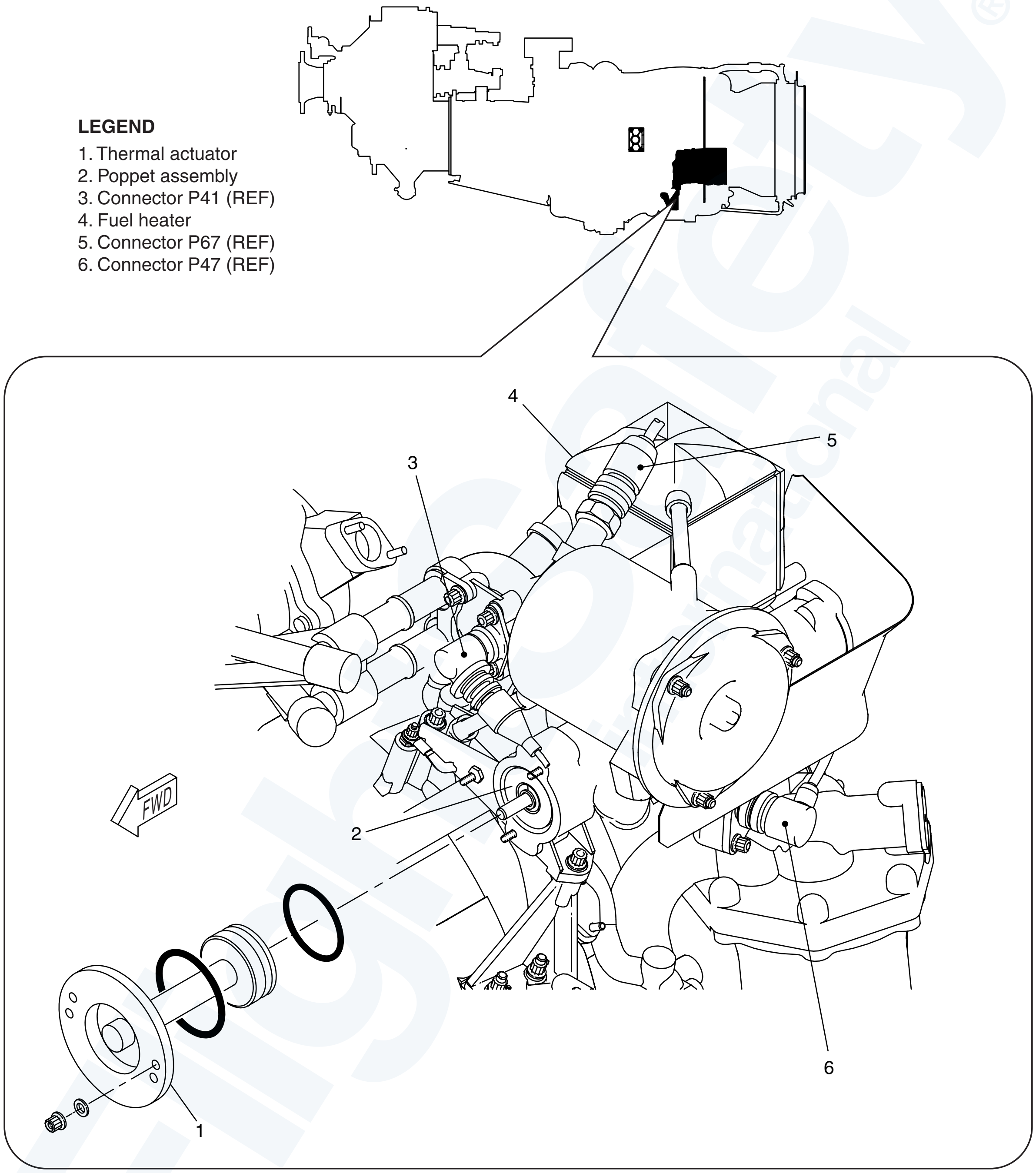

Figure 73-8. Fuel Heater Thermal Actuator Valve

Fuel Heater Thermal Actuator Valve

Refer to Figure 73-8. Fuel Heater Thermal Actuator Valve.

The thermal actuator valve is installed in the fuel heater.

The oil bypass valve is also equipped with a thermal sensor on the fuel side.

The thermal sensor keeps the valve fully opened when the fuel temperature is below 90°F (32°C).

As the fuel temperature increases above 90°F (32°C), the thermal sensor starts closing the oil bypass valve. When the fuel temperature reaches 120°F (49°C), the valve is fully closed and the oil bypasses the heater.

NOTES

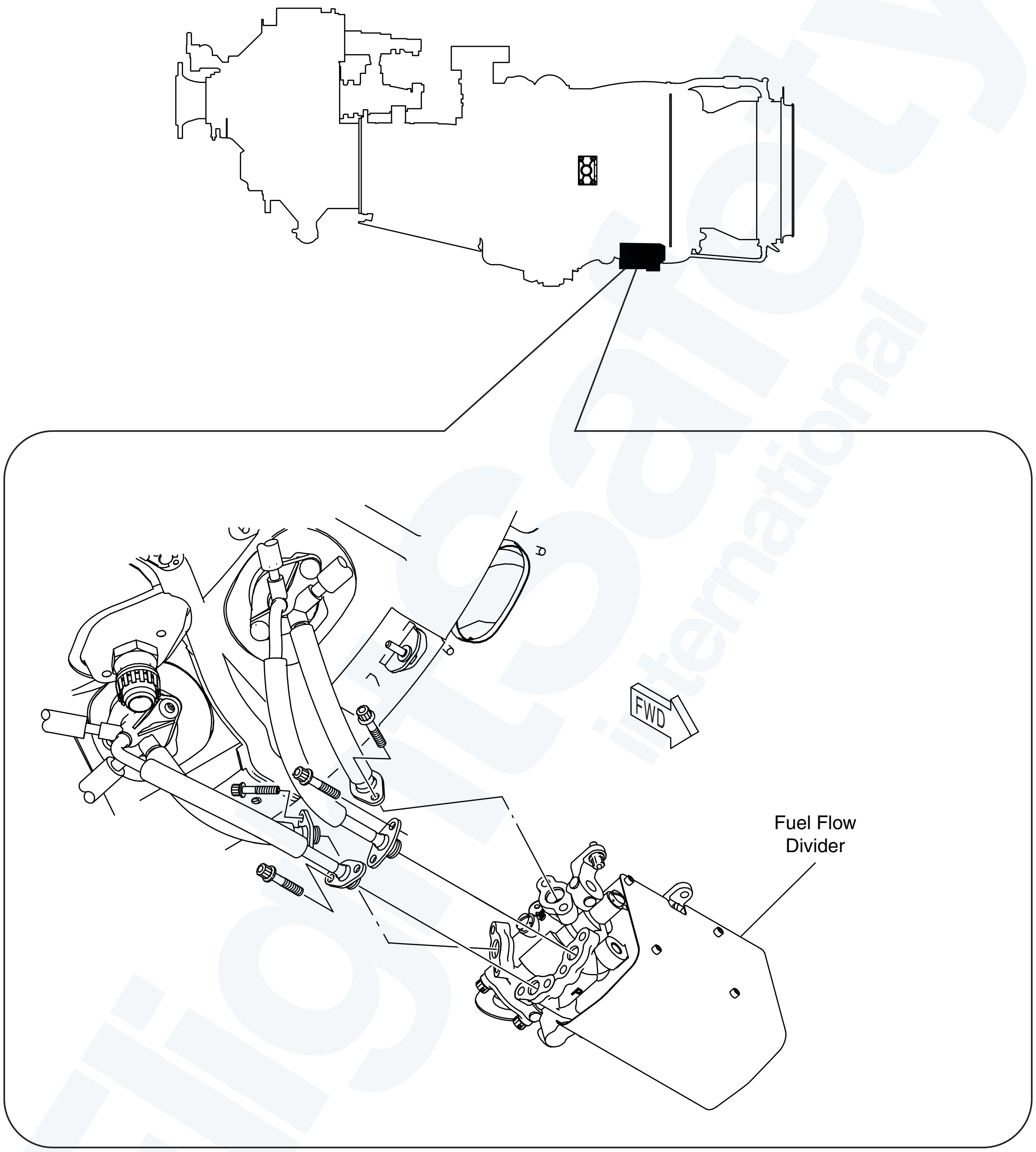

Figure 73-9. Flow Divider Valve 1

Flow Divider Valve (FDV)

NOTES

Refer to:

- Figure 73-9. Flow Divider Valve 1.
- Figure 73-10. Flow Divider Valve 2.

The FDV is installed on the bottom of turbine support case. The housing is made of cast aluminum. The purpose of the FDV is to divide the fuel flow between the primary and secondary fuel manifolds for starting and steady state operation.

During Start

Fuel from the FMU will flow to the FDV. At the FDV a pressure regulator is used to maintain a primary fuel pressure of 125 psi (862 kPa) above that of the secondary fuel pressure.

Fuel will then flow through the primary manifold and out the primary nozzle. As fuel pressure increases above the 125 psi differential fuel will now flow through the secondary manifold and out through the secondary nozzle

Steady State Operation

As fuel flow increases, the FDV equalizes the pressure between primary and secondary manifold. Equalization is maintained up to the maximum flow condition.

During Shutdown

Fuel left in the fuel manifolds is collected in an ecology reservoir integral to the FDV. This is done by the action of a piston and a spring.

Subsequent Start

Fuel stored in the ecology reservoir is also returned to the manifolds through the action of the ecology piston. A restricted orifice in the ecology circuit makes sure of a slow transition of the piston, so as not to affect the fuel nozzle operation during start. A fire shield is around the ecology reservoir for fire resistance purposes.

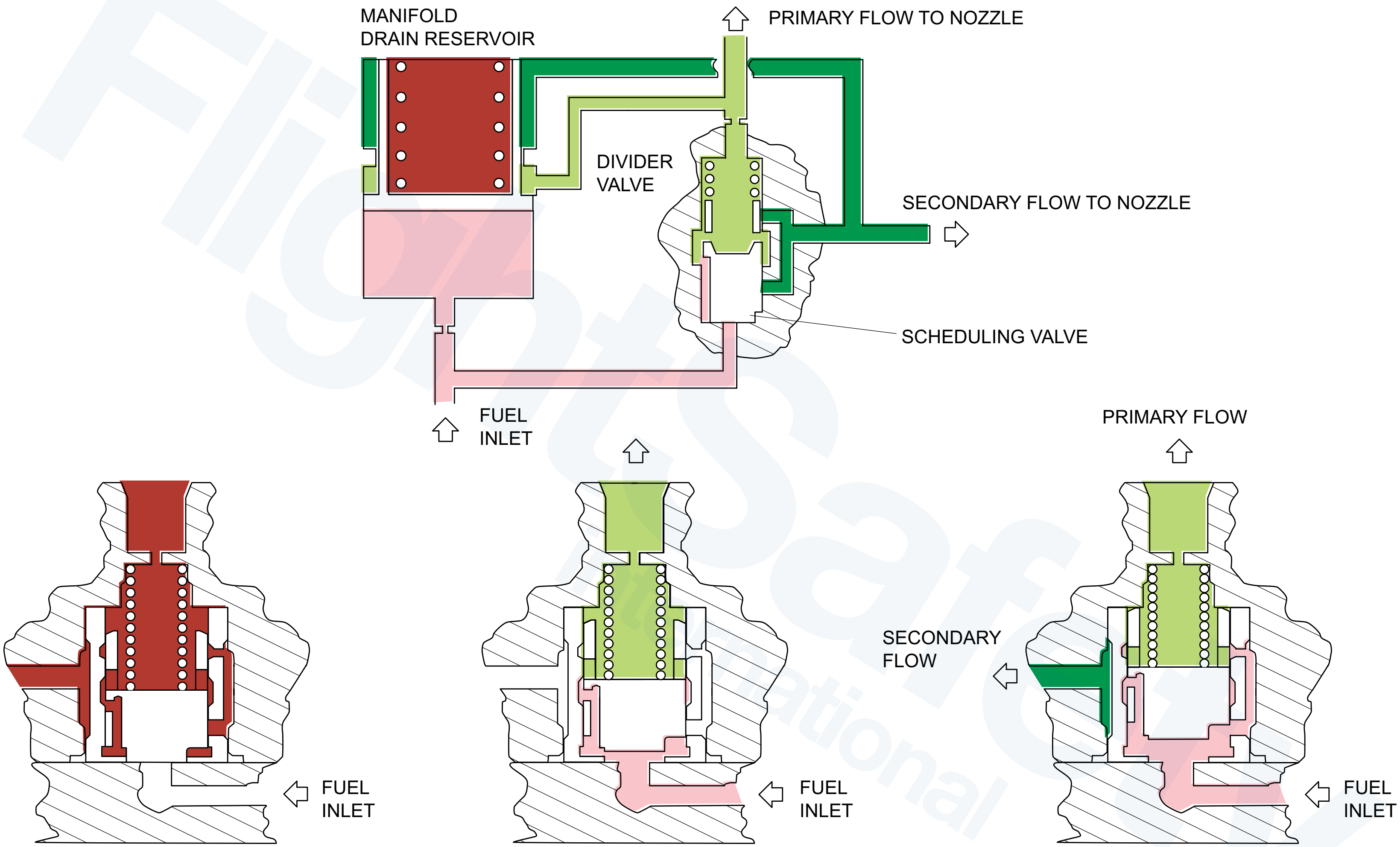

Figure 73-10. Flow Divider Valve 2

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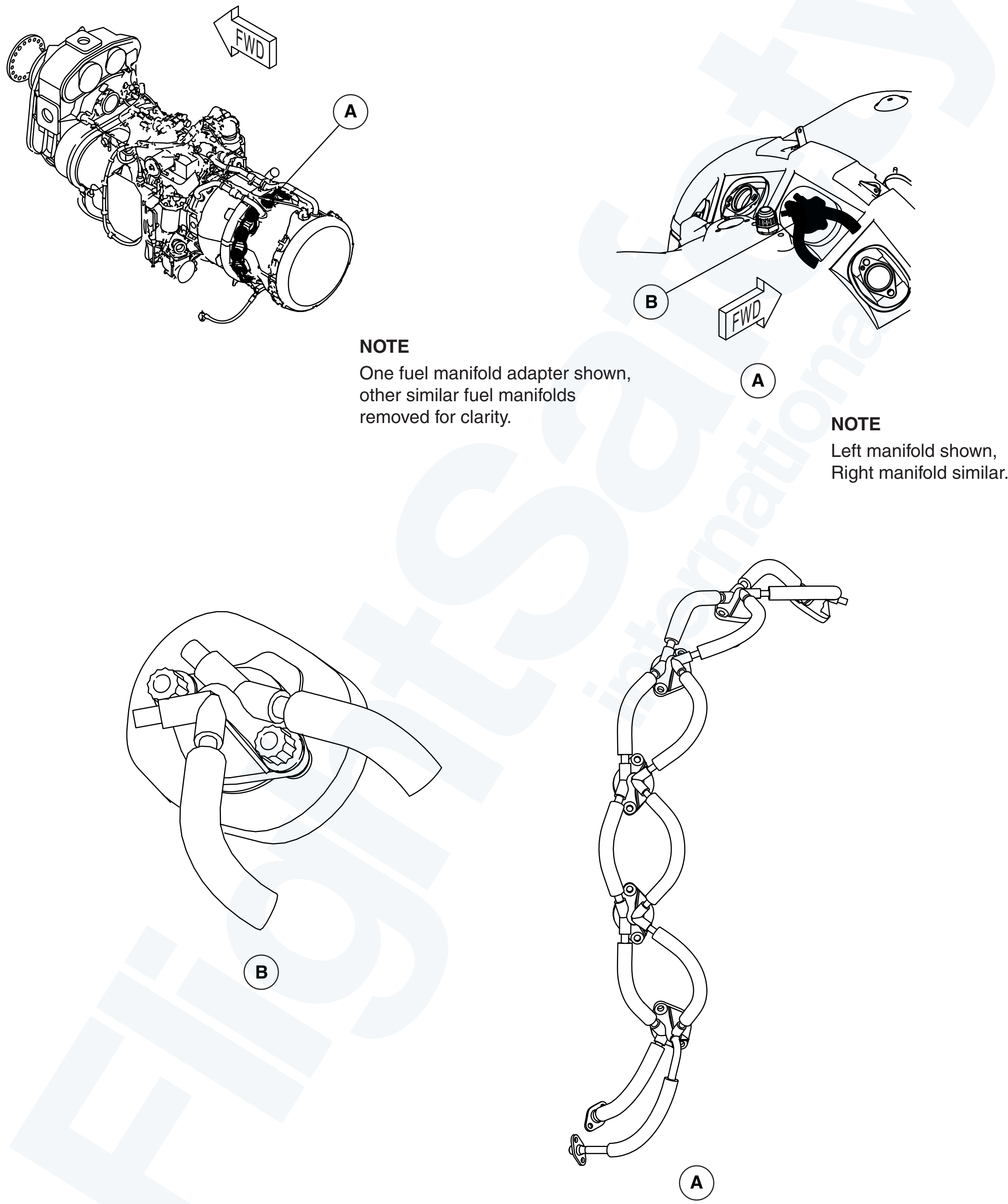

Figure 73-11. Fuel Nozzles and Manifolds

Fuel Nozzles and Manifolds

Refer to Figure 73-11. Fuel Nozzles and Manifolds.

The fuel manifolds are installed on the turbine support case; one on the left side and one on the right side, connecting the fuel nozzles together.

The fuel manifolds deliver fuel to the fuel nozzle adapters.

Each fuel manifold has two connections to the FDV; one for the primary fuel flow and one for the secondary fuel flow.

REMOVAL OF FUEL NOZZLES AND FUEL MANIFOLDS

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

- Operate the PULL FUEL/HYD handle to cut off the fuel
- The fuel manifold and fuel nozzles are removed as an assembly
- Ensure you have numbered the nozzles before removal for troubleshooting purposes
- Remove the nozzles from the manifold on a clean bench
- REMEMBER the nozzles between the spark igniters may be different subject to Service Bulletin compliance.

NOTES

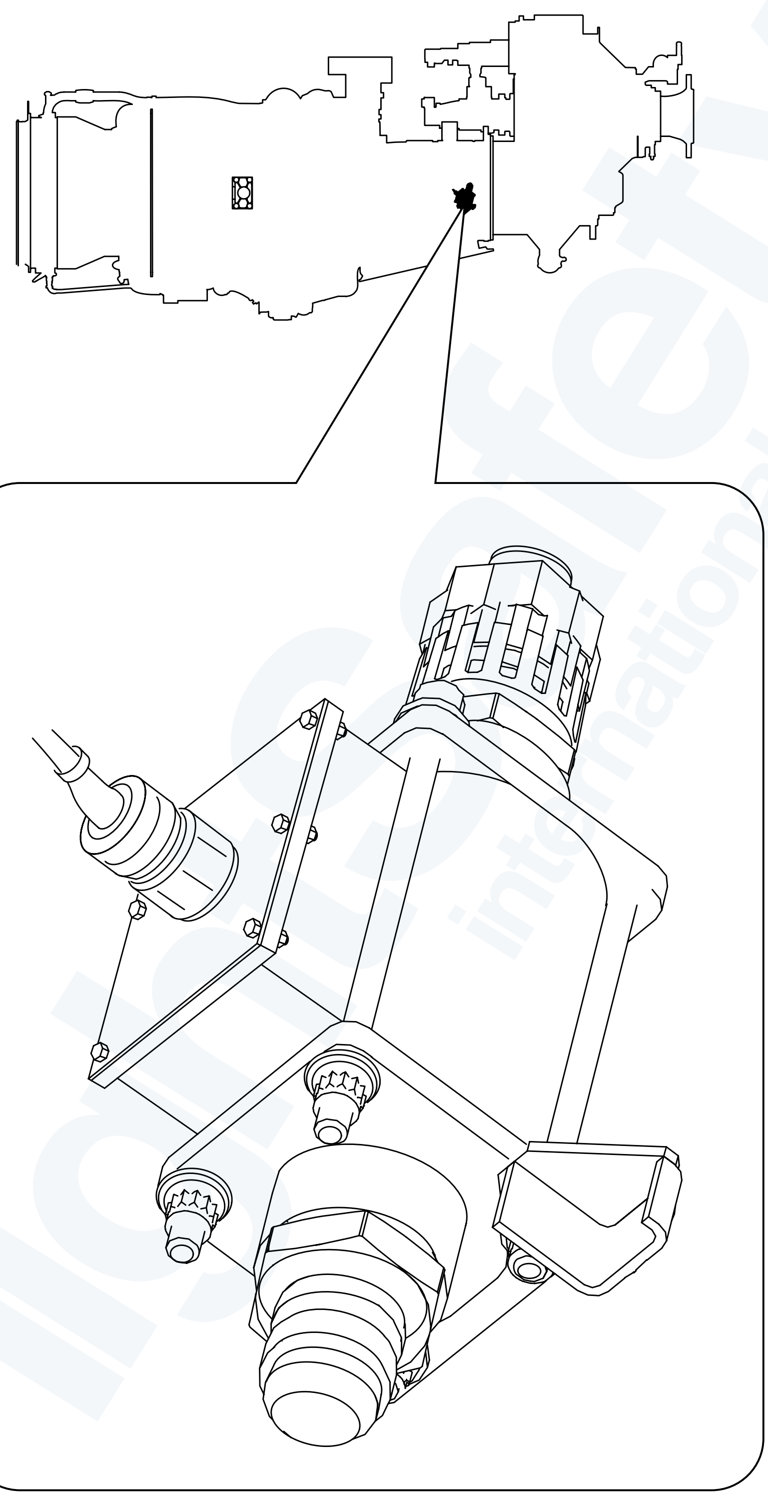

Figure 73-12. Fuel Flowmeter

Fuel Flowmeter and Fuel Tubes

Refer to:

- Figure 73-12. Fuel Flowmeter.
- Figure 73-13. Fuel Flowmeter and Tubes.

The fuel flow transmitter is on the right side of the engine intake section. The fuel flow transmitter converts fluid flow into an electrical signal and sends this information to the IFC.

The signal conditioning equipment in the IFC does the calculation to determine the mass flow rate of the fuel. This flow rate is shown on the ED.

The fuel flow transmitter is connected to the FMU on one side and to the fuel flow divider on the other; it is connected by two fuel tubes. The tubes have a braided flexible portion and a solid portion made of a titanium alloy.

The solid end of the tubes are attached to the flowmeter with a moeller-type nut. This is used for fire protection.

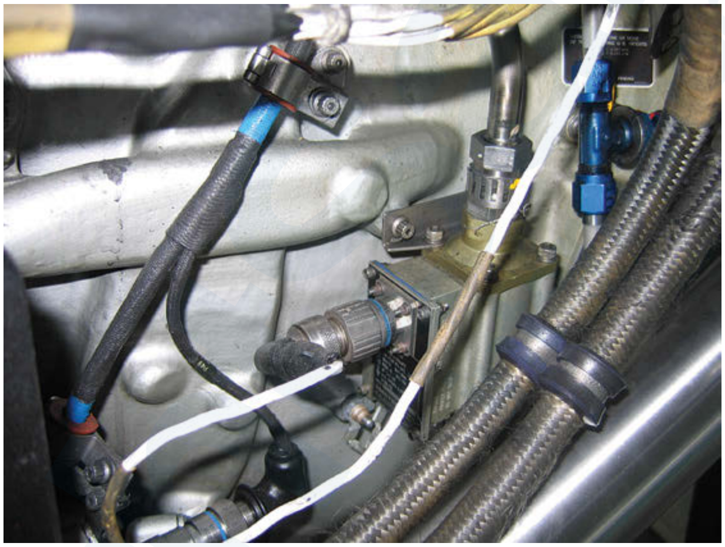

Figure 73-13. Fuel Flowmeter and Tubes

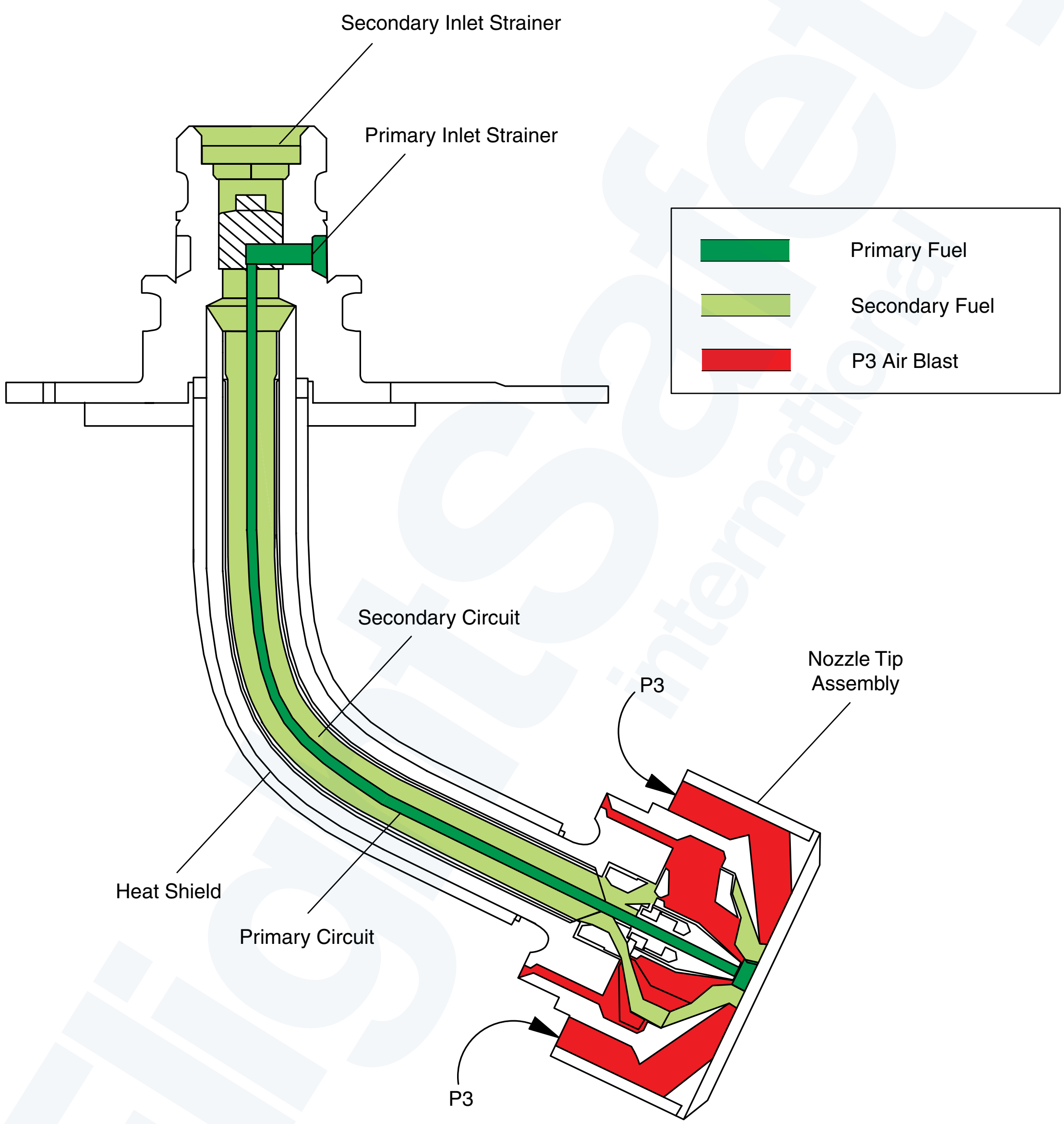

Figure 73-14. Fuel Nozzle Adapter

Fuel Nozzle Adapter

Refer to Figure 73-14. Fuel Nozzle Adapter.

The fuel nozzle adapters deliver fuel to the combustion chamber, where it is mixed with air and atomized for combustion.

There are twelve fuel nozzle adapters installed on the turbine support case.

The nozzle section of the adapter is inserted into the turbine support case. It is directed towards the front of the engine.

The nozzle protrudes into the combustion chamber through holes in the dome of the combustion chamber outer-liner. Fuel manifolds connect the fuel nozzles together.

All twelve fuel nozzle adapters are duplex units that have a primary and secondary orifice.

The primary fuel orifice is in the center of the nozzle; the secondary fuel orifices surround the primary orifice.

Use, for up to 1,000 hours of any restricted fuels that are listed in the AMM Chapter 12-10-28, necessitates a service check of the fuel nozzles.

NOTES

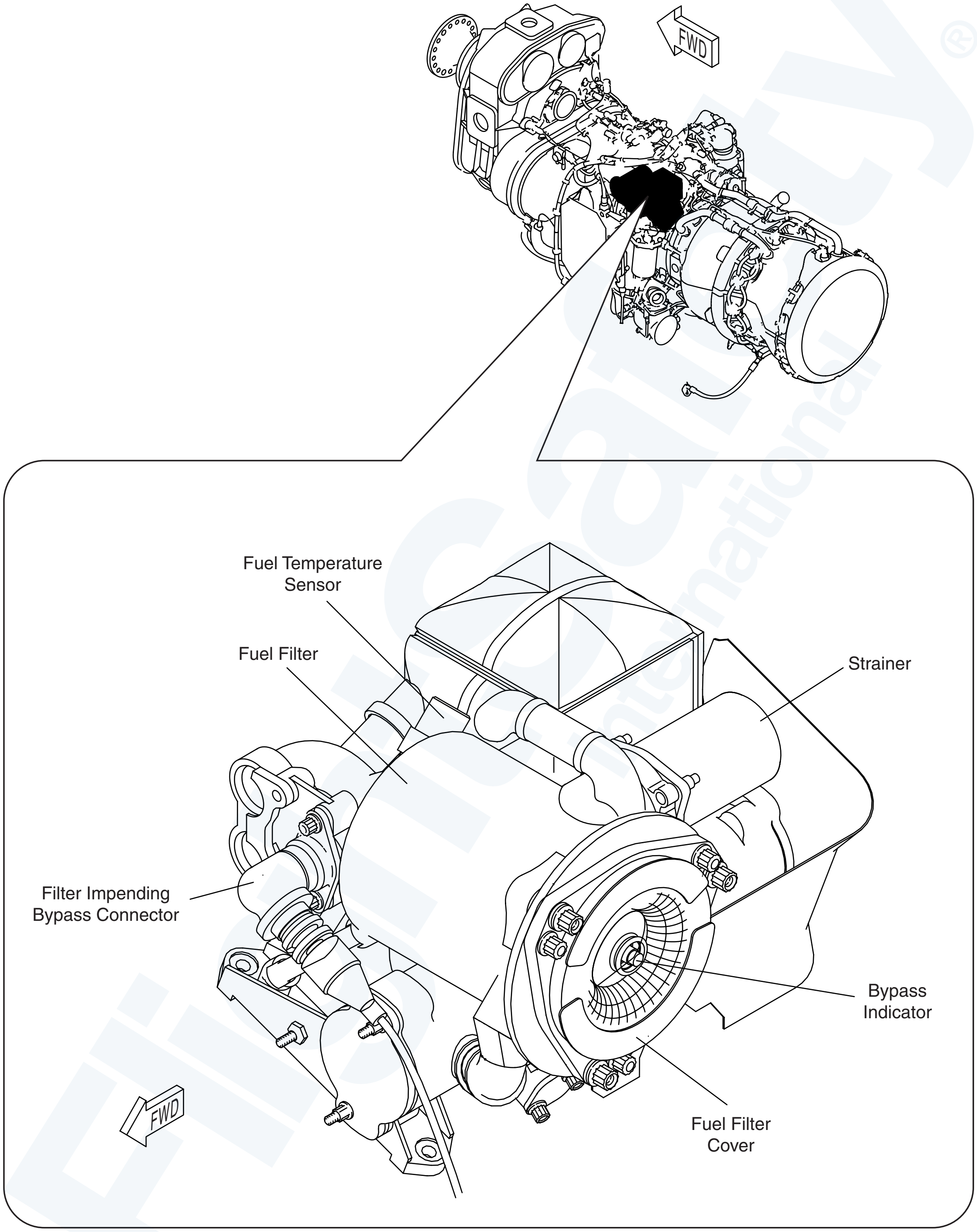

Figure 73-15. Fuel Filter

Fuel Filter

Refer to Figure 73-15. Fuel Filter.

The filter is downstream of the heat transfer matrix. It filters all fuel going to the engine.

Particles between 10 to 25 micron may pass through the filter. Any particles larger than 25 micron will be contained in the filter. The filter is protected by a bypass valve and a bypass indicator. The filter cannot be cleaned. It must be replaced.

NOTES

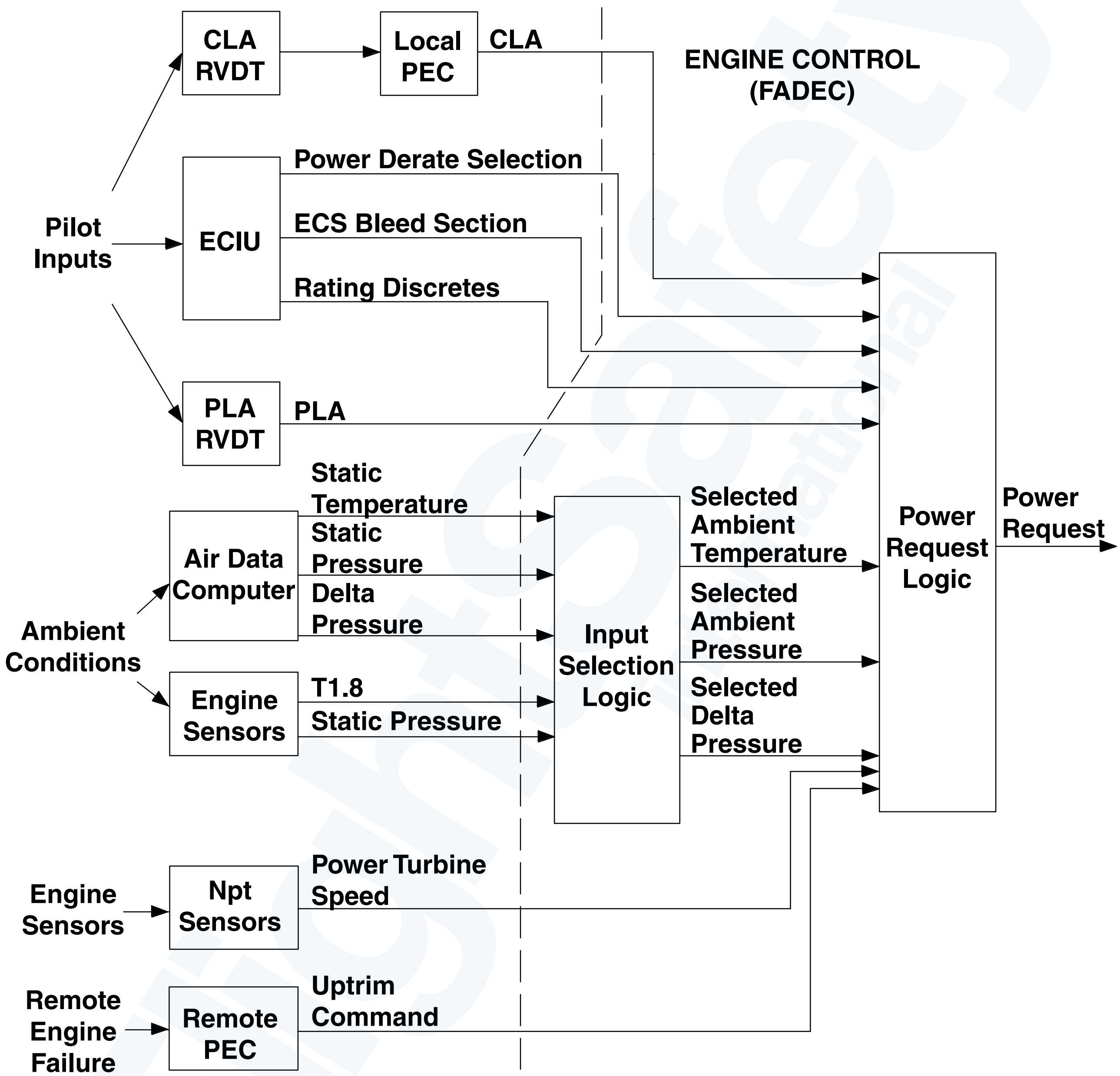

Figure 73-16. Power Request Block Diagram

73-21-00 ENGINE FUEL CONTROL SYSTEM

GENERAL

The engine fuel control system gives fuel at the required pressure and flow to control the engine power output.

SYSTEM DESCRIPTION

The engine fuel control system manages the powerplant by:

- Supplying fuel flow (scheduled as a function of the selected PLA)
- Monitoring engine ratings
- Measuring torque and ambient conditions.

The following components comprise this sub-system:

- FADEC
- FMU
- Fuel control electrical wiring harness
- Characterization plug and
- Permanent Magnet Alternator (PMA).

FADEC is a dual-channel microprocessor-based controller. One channel is in command, the other channel does indication. Normal channel change-over is achieved as follows:

- WOW
- Shutdown
- Next engine start completed.

The independent overspeed protection system uses redundant N_H signals from the engine to command a fuel shut off in the event of an N_H overspeed. The overspeed trip is set at 106% N_H. The overspeed circuitry is tested each time the engine is shut down.

The FADEC electronically controls the engine within safe thermal and mechanical operating limits.

FADEC does the following:

- Sends an electrical signal to the FMU to control engine power
- Controls the P2.7 HBOV and the P2.2 HBOV to prevent compression stall
- Prevents an engine overspeed
- Supervises engine starts and engine shutdowns by controlling the igniter exciter
- Detects and accommodates an Ni decouple
- Supplies voltage to the PEC when above 40% N_H speed)
- Detects and indicates faults
- Communicates with the ED
- Communicates with the Engine Monitoring Unit (EMU) in maintenance mode.
- Communicates to other units through the Universal Asynchronous Receiver Transmitter (UART) and ARINC data bus interfaces.

INSTALLATION OF EEC OF FADEC

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

- Install electrical connectors P1, P2, P40 and P50 ensuring the correct torque loading
- Install the characterization plug
- Install the EEC of FADEC ensuring bolts torque loaded correctly
- Close CBs in the order shown in the task
- Do a FADEC fault code clear
- Do a PEC fault code clear

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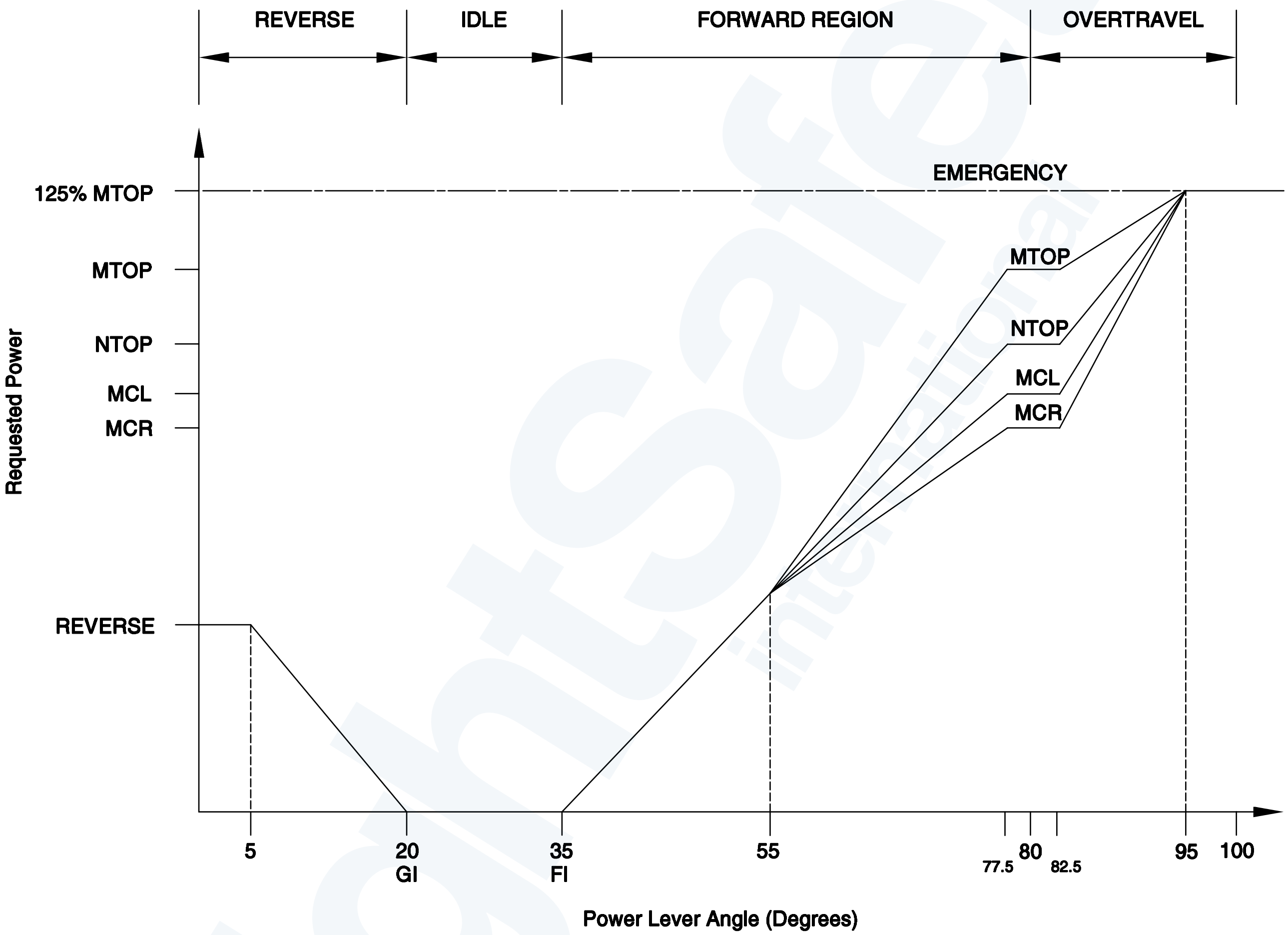

Figure 73-17. Power Lever Angles

Power Setting with the Power Lever

Refer to Figure 73-17. Power Lever Angles.

The power lever modulates power requests from Full Reverse (0°) to Rated Power Detent (77.5° to 82.5°).

Ground handling is achieved at any PLA below Flight Idle (35°).

Above 35°, the power request increases will be linear, with increasing PLA until the Rated Power Detent is reached.

Moving the power lever in the over-travel region (82.5° to 100°), increases requested power up to 125% of maximum take-off rating.

It also results in an increase of engine software limits. In this region the propeller control system automatically sets propeller speed to 1020 Np.

NOTES

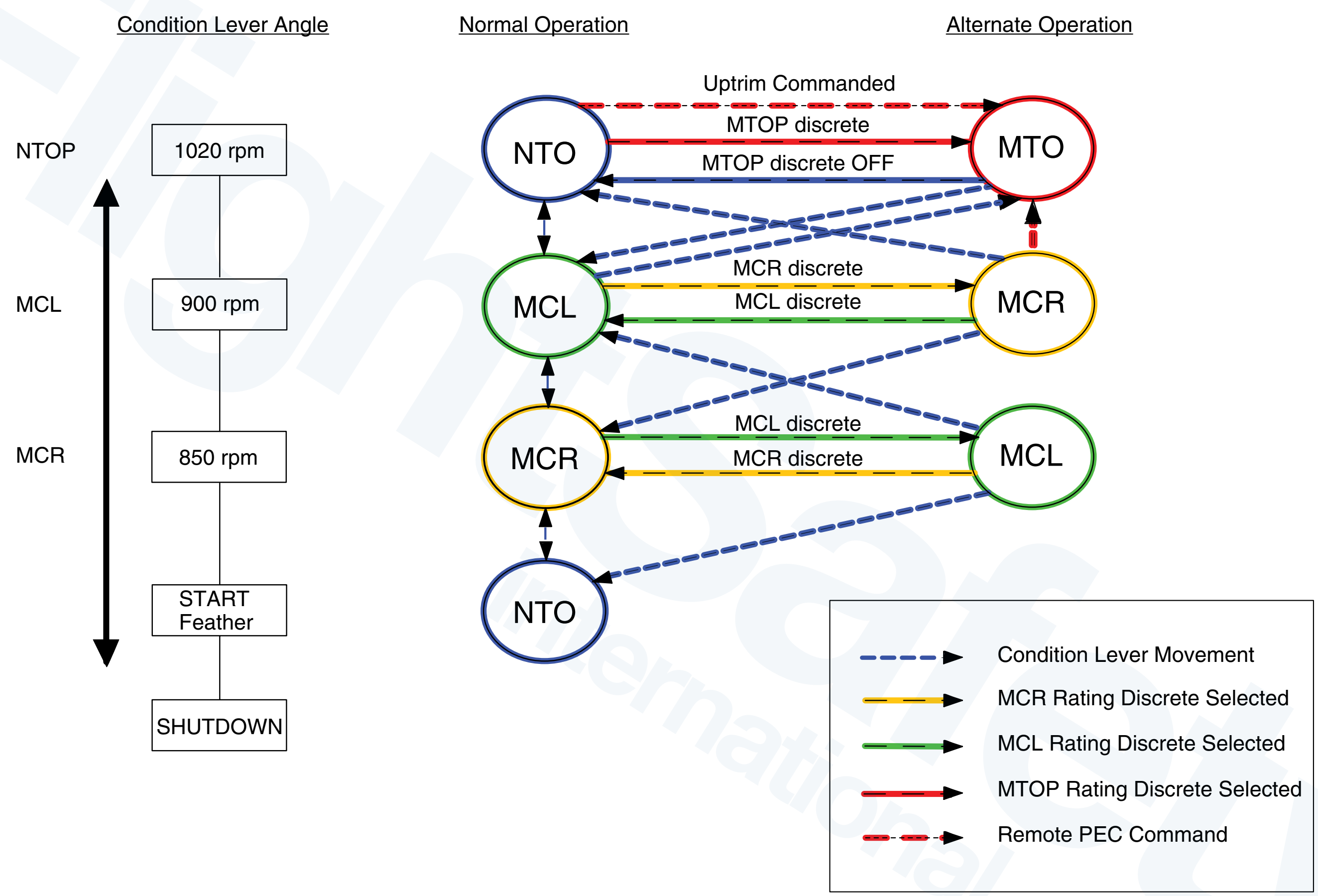

Figure 73-18. Alternate Power Rating Selections

Rating Selection with the Condition Lever

Refer to Figure 73-18. Alternate Power Rating Selections.

When the condition lever is moved to the detent positions, rating selection occurs at the same time with propeller speed selection.

The propeller control converts the Condition Lever Angle (CLA) to a position. It then transmits this position (as discrete information) to the FADEC (through the PEC/FADEC serial data buses).

For all condition lever positions, the Rated Power will be achieved when PLA is in the rating detent (77.5° to 82.5°).

When the PLA is reduced below the detent position, the power request for all positions converge to a single point at 55°.

In the PLA overtravel region, the power request for all ratings converge to 125% maximum power at 95°.

Selection of Alternate Power Rating and Propeller Speed Combinations (Rating Discretes)

Alternate combinations of propeller speed and engine power rating can be set by using the Maximum Take-Off Power (MTOP), Maximum Climb (MCL) and Maximum Cruise (MCR) rating discretes (in the flight compartment).

These discretes are transmitted to the FADEC through the Engine Cockpit Interface Unit (ECIU). They change the rating normally selected as a function of CLA under certain conditions.

The NTOP is selected by FADEC whenever the condition lever is in the 1020 RPM position.

The MTOP rating is defined as the maximum available power certified for take-off operation. The MTOP switch is an alternate action switch.

When CLA is in the 900 RPM position, and the MCR discrete is selected, the MCL rating normally associated with this propeller speed is overridden by the MCR rating.

The MCR rating discrete is a momentary action switch, so any movement of the CLA will base engine rating selection on the new CLA position.

Alternatively, the MCL rating can be recovered, at the same CLA position, by selecting the MCL discrete. The MCL discrete is also a momentary action switch.

Selection of MCL at a CLA of 850 RPM is also possible using the MCL rating discrete.

Environmental Control System (ECS) Bleed Selection

Refer to Table 73-1. ECS Bleed Selection.

The power requested (at a given power rating) is a function of the selected ECS bleed when the engine is operating at the thermal limit. The ECS bleed selection is translated in bleed levels, used for thermal power rating calculations. Higher amounts of ECS bleed results in less thermal rated power. This may reduce the power requested for a given power rating, depending on the ambient conditions.

The FADEC discriminates between single and dual ECS bleed by using the following logic. Dual engine level is used unless the power rating is MTOP, demanded by Uptrim only, or ECS is selected OFF on the remote engine.

The ECS bleed selection is also used by FADEC to distinguish between MTOP and Maximum Continuous Power (MCP).

BLEED	MTOP/MCP
OFF	MTOP
MIN	MTOP
NORMAL	MCP
MAX	MCP

Table 73-1. ECS Bleed Selection

Q400 POWER CALCULATIONS
SHP = Tq X Np X K
5071 (MTOP) = 100% Tq X 1020 Np X K
K = 5071 / 100 X 1020   K = 0.04972
Tq = SHP / Np X K

NTOP (Tq) =	4580 / 1020 X 0.04972	NTOP = 90% Tq
MCL (Tq) =	4058 / 900 X 0.04972	MCL = 91% Tq
MCR/900 (Tq) = DISCRETE	3947 / 900 X 0.04972	MCR/900 = 88% Tq
MCR (Tq) =	3947 / 850 X 0.04972	MCR = 93% Tq
MCL/850 (Tq) = DISCRETE	4058 / 850 X 0.04972	MCL/850 = 96% Tq
MAX REV (Tq) =	1500 / 950 X 0.04972	MAX REV = 32% Tq

Table 73-2. Power Derate Selection

Power Derate Selection

Refer to Table 73-2. Power Derate Selection.

The power can be reduced for take-off in the NTOP rating using the power derate function. To derate the requested power, the power derate discrete is pressed, with the CLA at the 1020 RPM position and PLA below the rated power detent. Selection of the power derate discrete momentary switch decreases the NTOP requested in steps of 2%, to a limit of 10%. Selection of the power derate reset discrete, at any time, resets the derate to 0%.

A power derate is permitted only when both FADEC channels of each powerplant have received a power derate command.

Confirmation occurs through cross powerplant communication.

The power derate function cannot be activated in MTOP or MCP rating. If an Uptrim is commanded from the remote powerplant, the requested derate will apply to the MTOP requested power.

Automatic Take-Off Power Control System

The Automatic Take-off Power Control System (ATPCS) increases the power of an engine if the other engine loses power. This is referred to as Uptrim, or, Automatic Take-Off Thrust Control System (ATTCS).

The local engine FADEC will respond to the Uptrim signal from the remote Propeller Electronic Control/Autofeather (PEC/AF) unit by changing from NTOP to MTOP/MCP.

The ATPCS is active during take-off and go around maneuvers. The local ATPCS is armed when:

- Both local and remote PLA are high
- Local engine torque is high.

If the local engine fails (indicated by low torque), an Uptrim signal is sent by the local PEC to the remote FADEC. An Uptrim condition is shown in the flight compartment by:

- Uptrim indication
- Change in engine rating from NTOP to MTOP/MCP
- Change in the torque bug from NTOP to MTOP/MCP%.

Mechanical and Thermal Power Limits

Refer to Table 73-3. Power Limit.

The engine power limit logic is the lowest value between the mechanical power limit and the thermal power limit (for the selected rating).

The mechanical power limit is set as a function of the engine rating.

Rating Selected	Mechanical Power SHP
MTOP/MCP	5071
NTOP	4580
MCL	4058
MCR	3947

Table 73-3. Power Limit

The thermal power rating is set as a function of the following conditions:

- Rating selected
- Ambient temperature
- Aircraft altitude
- Aircraft speed
- ECS bleed air extraction
- Power turbine shaft speed.

Figure 73-19. Fuel Metering Unit

COMPONENT DESCRIPTION

Fuel Metering Unit

Refer to Figure 73-19. Fuel Metering Unit.

The FMU is on the top of the compressor section. It controls the fuel flow to the engine (based on demand) through the FADEC. The FADEC calculates the amount of fuel to supply based on N_H and Np.

The FMU is installed on the Permanent Magnet Alternator (PMA) on the LP compressor case. It is attached with a V-band clamp.

The FMU has two integral fuel pumps: a low pressure pump and a high pressure pump. Both pumps are engine-driven.

The FMU has a fuel inlet port connected to the airframe fuel supply and a fuel outlet port connected to the fuel flowmeter. It also has fuel heater inlet and outlet ports. The FMU has electrical connections to the EEC and airframe.

The FMU modulates the engine fuel flow over the entire operational envelope of the engine. It does this in response to signals sent by the FADEC.

Fuel from the low pressure pump is routed to the fuel oil heater. From the fuel heater, fuel is then routed to the inlet of the high pressure pump. From there, it enters the metering portion of the FMU.

The FMU is composed of the following components:

- Servo pressure regulator
- Metering valve
- Metering valve torque motor
- Dual coil LVDT
- PRV/MFC
- Minimum Pressure and Shut-Off Valve (MPSOV):
  - High pressure relief valve
  - Dual overspeed and shutdown solenoids Vapor venting check valve.

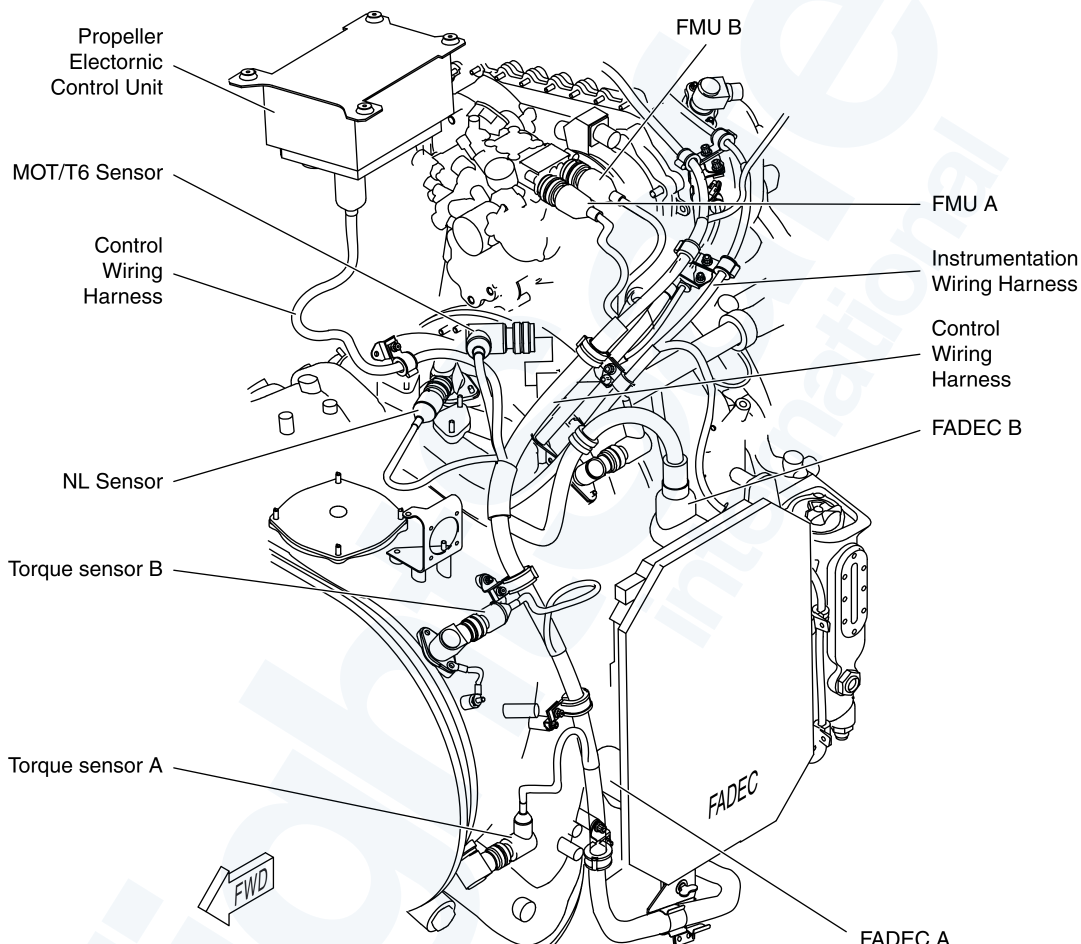

Figure 73-20. Electrical Wiring - Fuel Control Harness (1 of 3)

Electrical Wiring - Fuel Control Harness

Refer to:

- Figure 73-20. Electrical Wiring - Fuel Control Harness (1 of 3).
- Figure 73-21. Electrical Wiring - Fuel Control Harness (2 of 3).
- Figure 73-22. Electrical Wiring - Fuel Control Harness (3 of 3).

The fuel control electrical wiring harness connects the sensors to the FMU, and the PEC to the FADEC.

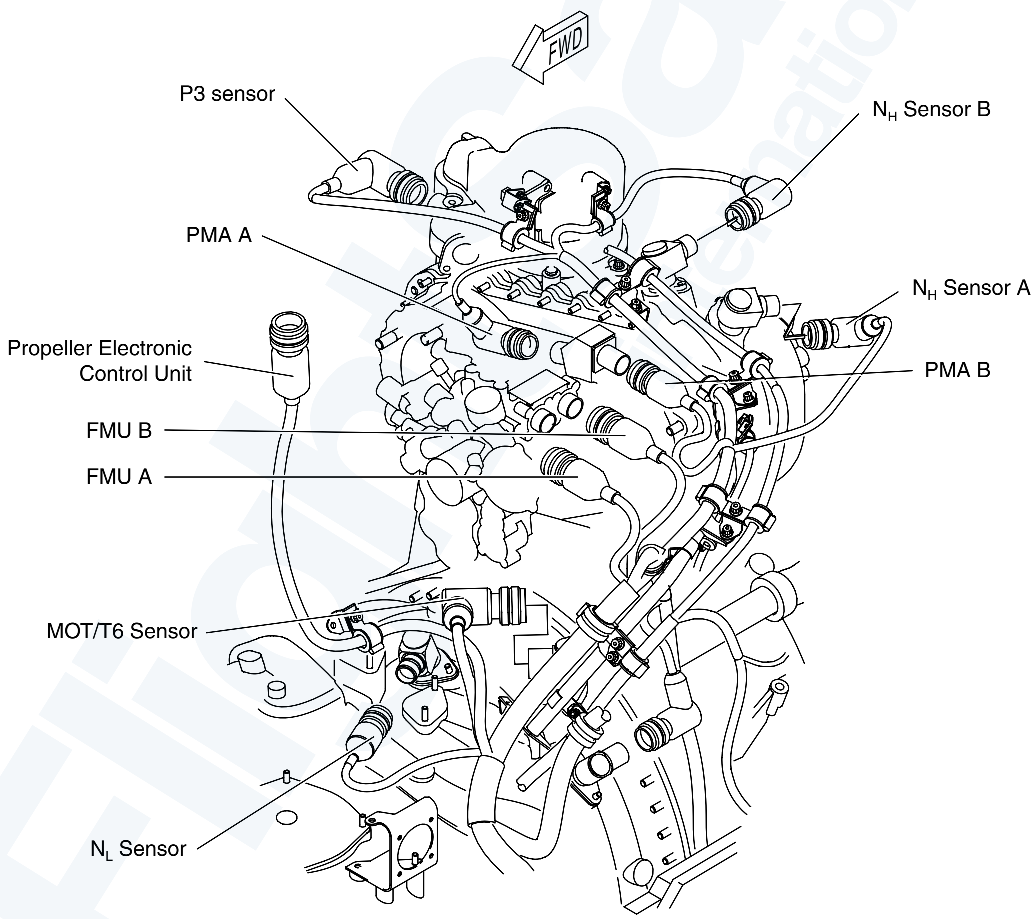

Figure 73-21. Electrical Wiring - Fuel Control Harness (2 of 3)

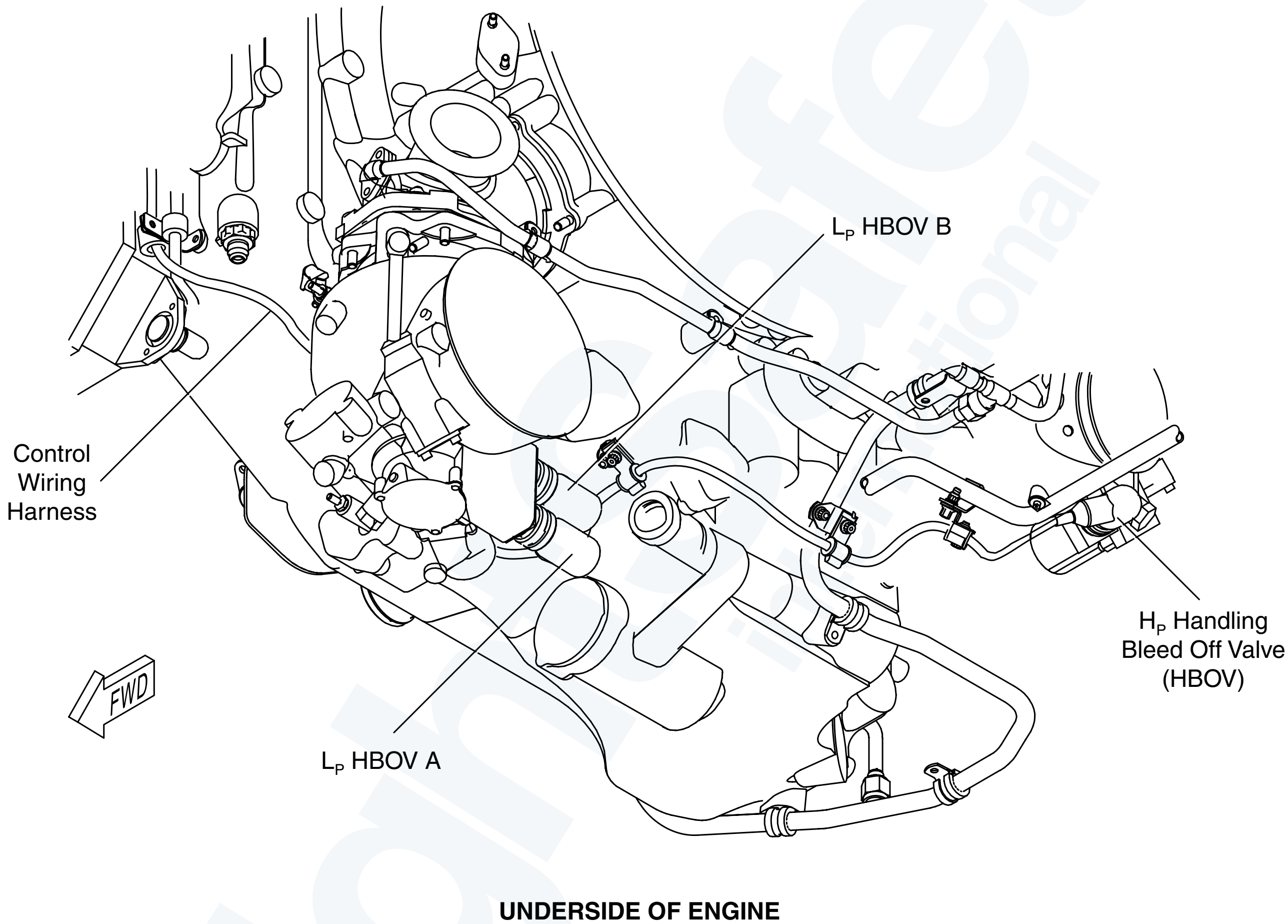

Figure 73-22. Electrical Wiring - Fuel Control Harness (3 of 3)

Permanent Magnet Alternator

Refer to Figure 73-23. Permanent Magnet Alternator.

The PMA is an integral part of the FMU and is driven by the Accessory Gearbox (AGB). The PMA features a rotor and a stator. The rotor is supported by the AGB bearings and the fuel pump bushings.

The PMA supplies electrical power to the FADEC and the PEC.

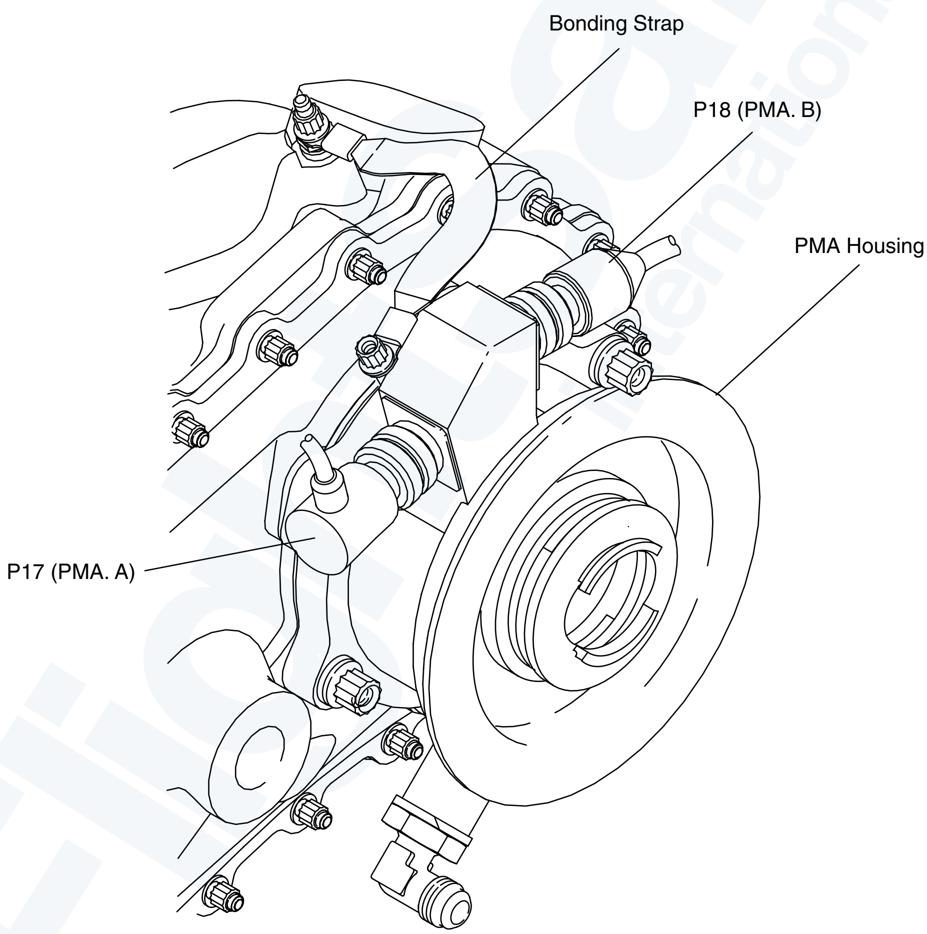

Figure 73-23. Permanent Magnet Alternator

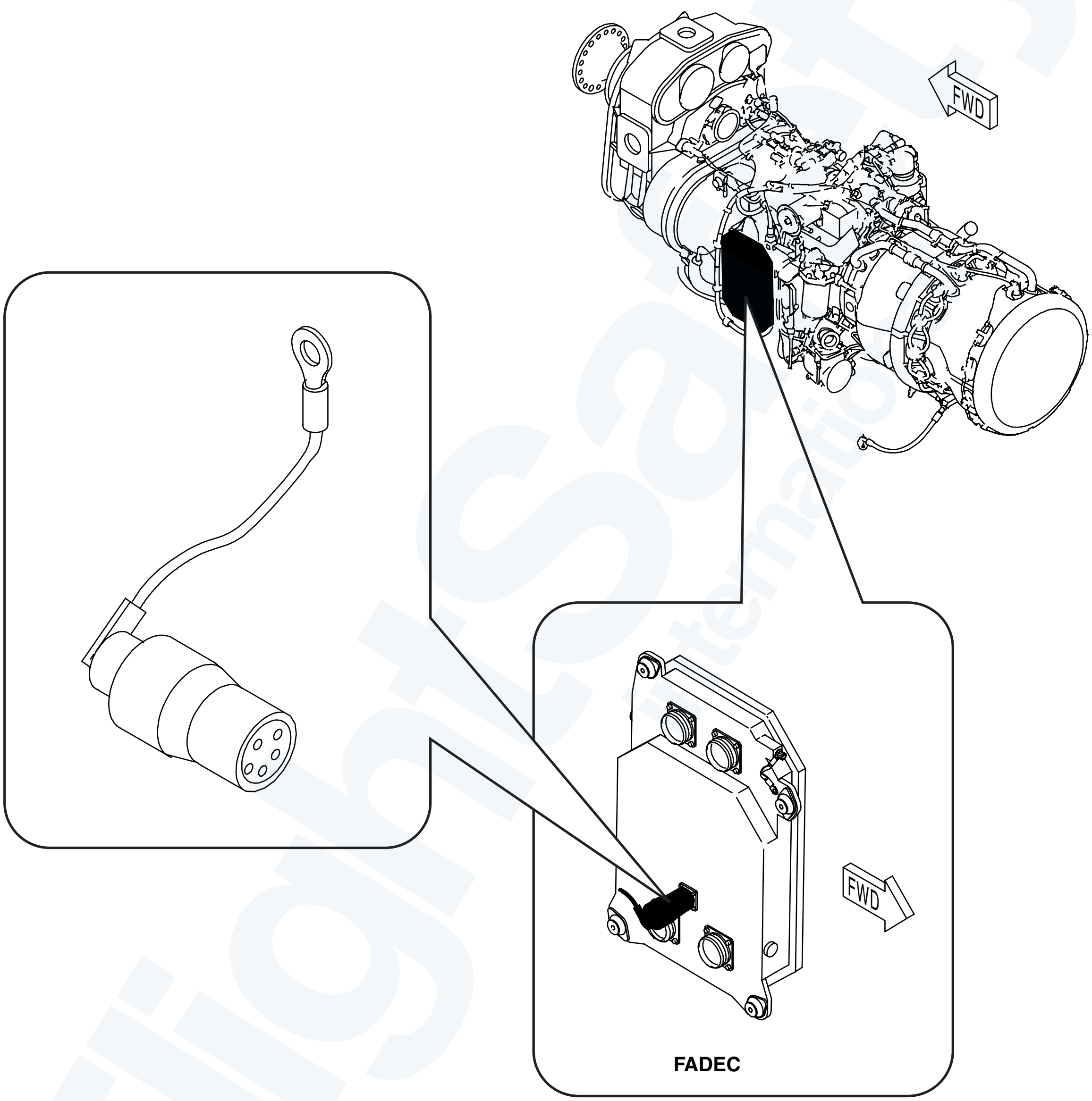

Figure 73-24. Characterization Plug

Characterization Plug

Refer to:

- Figure 73-24. Characterization Plug.
- Figure 73-25. Characterization Plug Side View.
- Figure 73-26. Characterization Plug Top View.

The characterization plug is installed on the FADEC at the J3 connector. The plug is physically connected to the FADEC channel A only. The trim values are passed to channel B by the FADEC internal communication bus. The plug is attached to the turbomachinery with a lanyard.

Inside the plug there are connections for four resistors. Two are for torque calculations, used for the torque-shaft gain (slope) and bias (offset) trim values, and a third one is used for the engine model identification. The fourth resistor location is unused.

The value of each resistor is determined during the engine test and can only be done by a qualified overhaul facility. A sealing compound is put on the resistors after their installation in the plug.

When assembled, a torque-shaft almost always has differences in the spacing of the teeth used to generate a signal for the torque sensor to read. These physical spacing differences require electrical trimming for a correct torque indication. The bias trim resistor is used to compensate for these differences.

The gain resistor compensates for the material characteristics of the torque-shaft. These characteristics include things such as the effect of temperature on the metal elasticity of the torque-shaft material.

The characterization plug can only be replaced with a plug of the same class. The trim values of the plug are marked on the RGB data plate.

Characterization Plug Installation

- Check the resistance values of the plug by reading them on the decal on side of the plug
- Compare the values to those on the Engine Reference Data Plate
- The values must be the same
- Install the plug to FADEC
- Install FADEC
- Connect retaining strap of the plug to flange C on the engine
- Do a PLA trim
- Do a FADEC fault code clear.

Figure 73-25. Characterization Plug Side View

Figure 73-26. Characterization Plug Top View

73 FUEL

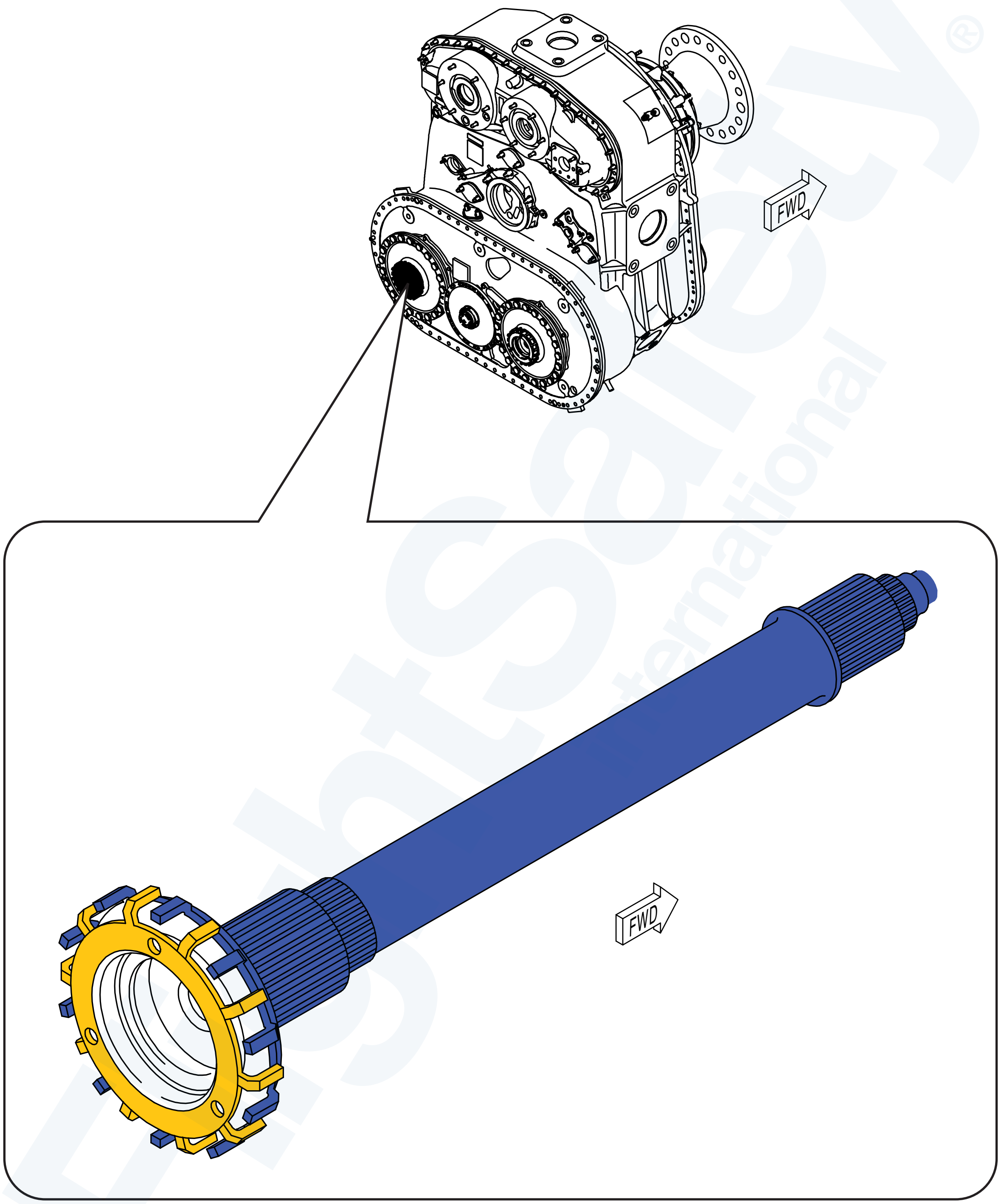

Figure 73-27. Torqueshaft

Torqueshafts

NOTES

Refer to Figure 73-27. Torqueshaft.

Torqueshafts have variations in the static spacing of the teeth (due to manufacturing tolerances). These teeth generate a signal that the torque sensor reads. These physical spacing differences require electrical trimming before a correct torque reading can be made.

The bias trim resistor compensates for these differences. It is also necessary to compensate for the differences in individual torque shaft stiffness (due to material and manufacturing tolerances).

The gain resistor compensates for this.

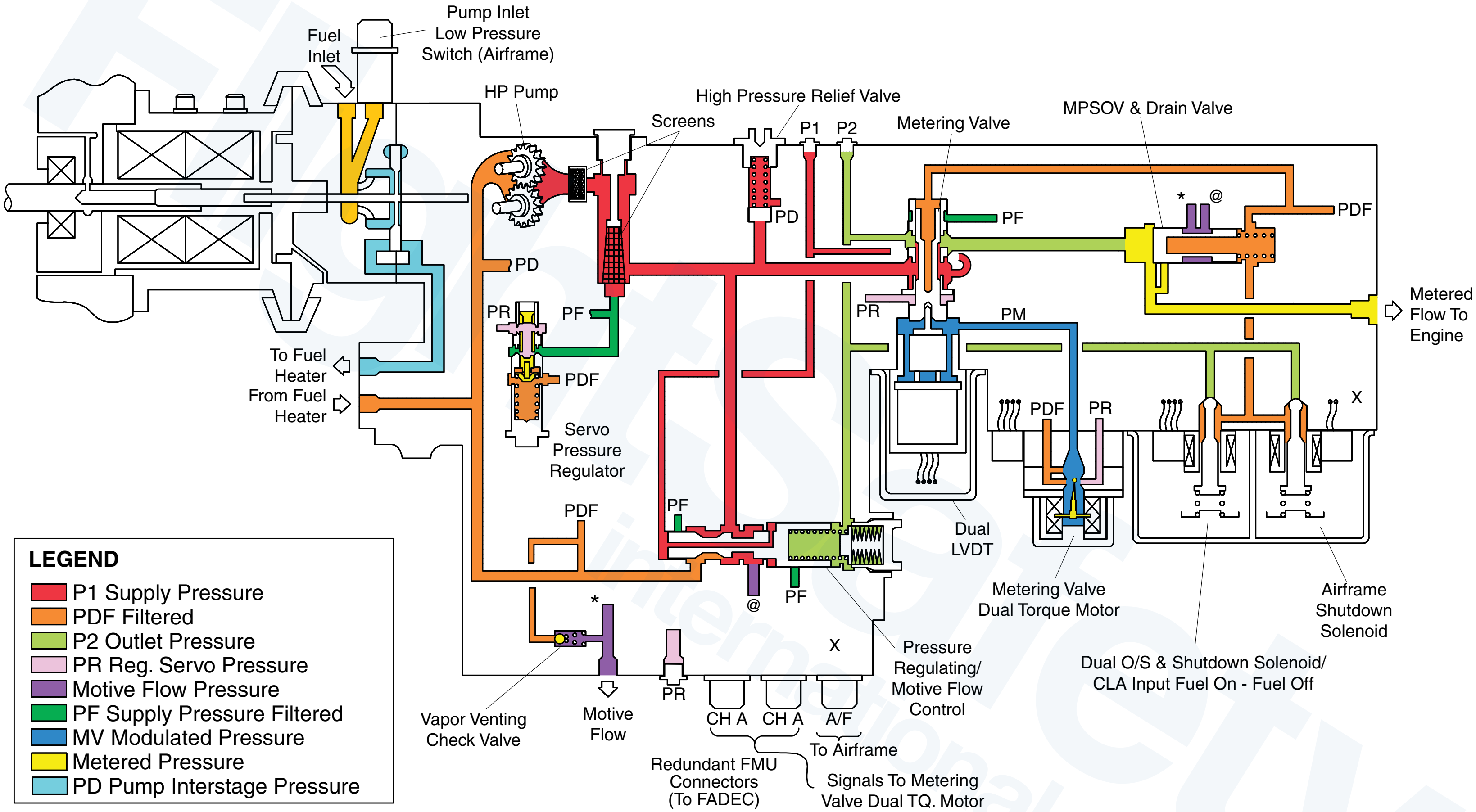

Figure 73-28. Fuel Metering Unit Schematic

OPERATION

Normal System Operation

Refer to Figure 73-28. Fuel Metering Unit Schematic.

The pump gears of the FMU are driven by a spline drive shaft (at accessory gearbox speed).

A quill shaft (between the gearbox end spline and the pump end spline) allows for misalignment. A carbon face seal prevents fuel from entering the pump/PMA interface cavity. Any fuel entering the cavity is drained overboard.

Pump drive spline lubrication is provided from the AGB.

Oil return channels in the PMA stator housing return the oil to the gearbox.

The metering valve is a half area type servo valve. Regulated servo pressure (PR), acting on a half-area servo piston, is balanced by a modulated pressure acting on the valve area.

The pressure is modulated by the dual coil torque motor.

The servo pressure regulator:

- Supplies the metering valve with a fine filtered pressure source

The metering valve:

- Controls the fuel flow to the engine
- Fuel flow (Wf) can vary between 125 pph to 2875 pph
- Bias to close.

The current input to the torque motor modulates the flapper, thereby allowing fuel to flow either into or out of the metering valve piston end, depending on direction of flapper motion. As the metering valve position changes, the fuel flow metering window area changes with a known relationship.

The metering valve torque motor:

- Controls the metering valve modulated pressure (PM) to operate the metering valve (up and down movement)
- Operated by the FADEC channel A and B
- The torque motor is designed with a null bias to close Wf with loss of electrical power.

The dual coil LVDT:

- Detects metering valve position
- Attached to the valve spool
- Electrical position is utilized by the FADEC to close the minor loop around the metering valve servo.

The PRV/MFC splits pump excess flow into two paths:

- To the motive flow line and then to the tank ejector pumps
- To the pump interstage (pd)
- The PRV will regulate a fixed pressure of 40 psid across the metering valve window and PRV damping orifice.

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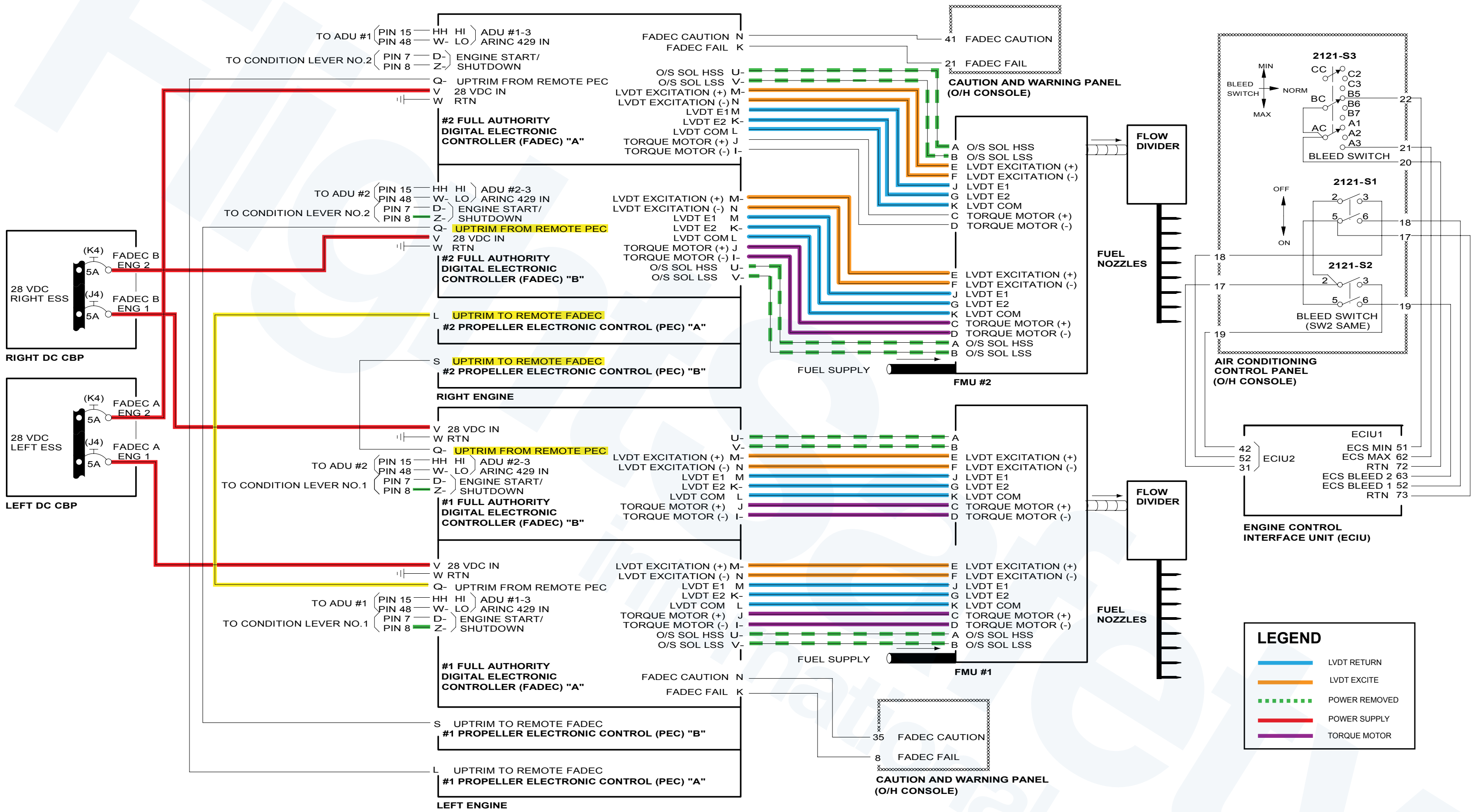


Figure 73-29. FADEC Control

Refer to Figure 73-29. FADEC Control.

The Minimum Pressure and Shut Off Valve (MPSOV):

- Provides a minimum control inlet pressure for proper operation
- Maintains a minimum fuel pressure (250 to 275 psid) in the FMU during low flow condition
- Assists the dual overspeed and shutdown solenoid to start and shutdown the engine
- Allows the motive flow fuel to supply the motive flow pumps in the system (engine and airframe).

The Pressure Regulating/Motive Flow Control Valve (PRVMFCV):

- Directs motive flow fuel to MPSOV. Opens at about 10% N_H
- Maintains pressure differential of 40 psid between MV inlet pressure (P1) and MV outlet pressure (P2)
- The pressure difference is changed with fuel temperature by bimetallic discs acting on the PRV spring
- This keeps the metering valve mass flow constant at a fixed valve position regardless of the fuel density change due to temperature
- Any additional pump excess flow is now bypassed to Pd.

The high pressure relief valve:

- Prevents excessive pressure in the FMU
- Opens at 1450 ± 50 psid (9997 ± 345 kPad).

The dual overspeed and shutdown solenoids:

- Are provided to shutoff metered flow to the engine when energized by the FADEC for either normal shutdown, or overspeed
- A second independent solenoid, allows the pilot to shutdown fuel flow to the engine. It is operated by the pull fuel/hyd handle (airframe)
- Pressure is ported to the spring side of the MPSOV when either solenoid is opened. The higher pressure plus spring force closes the MPSOV.

Vapor venting check valve:

- Vents vapor when the aircraft boost pump is initiated (prior to starting the engine)
- Prevents vapor lock in the FMU that could affect the operation
- Opens at 2 to 9 psid (14–62 kPa)
- Vapor will be returned to the tank through the motive flow line.

Pump inlet low pressure switch (airframe):

- Indicates the FADEC, ECIU, CAWS and EMU that the low pressure fuel has dropped below 5.5 ± 0.8 psi (38 ± 5.5 kPa).

Electrical power to the EEC of FADEC is from the aircraft essential buses.

Power from the right essential bus is supplied to Pin V of the Channel B for FADEC No.1 and No.2.

Power from the left essential bus is supplied to Pin V of the Channel A for FADEC No.1 and No.2.

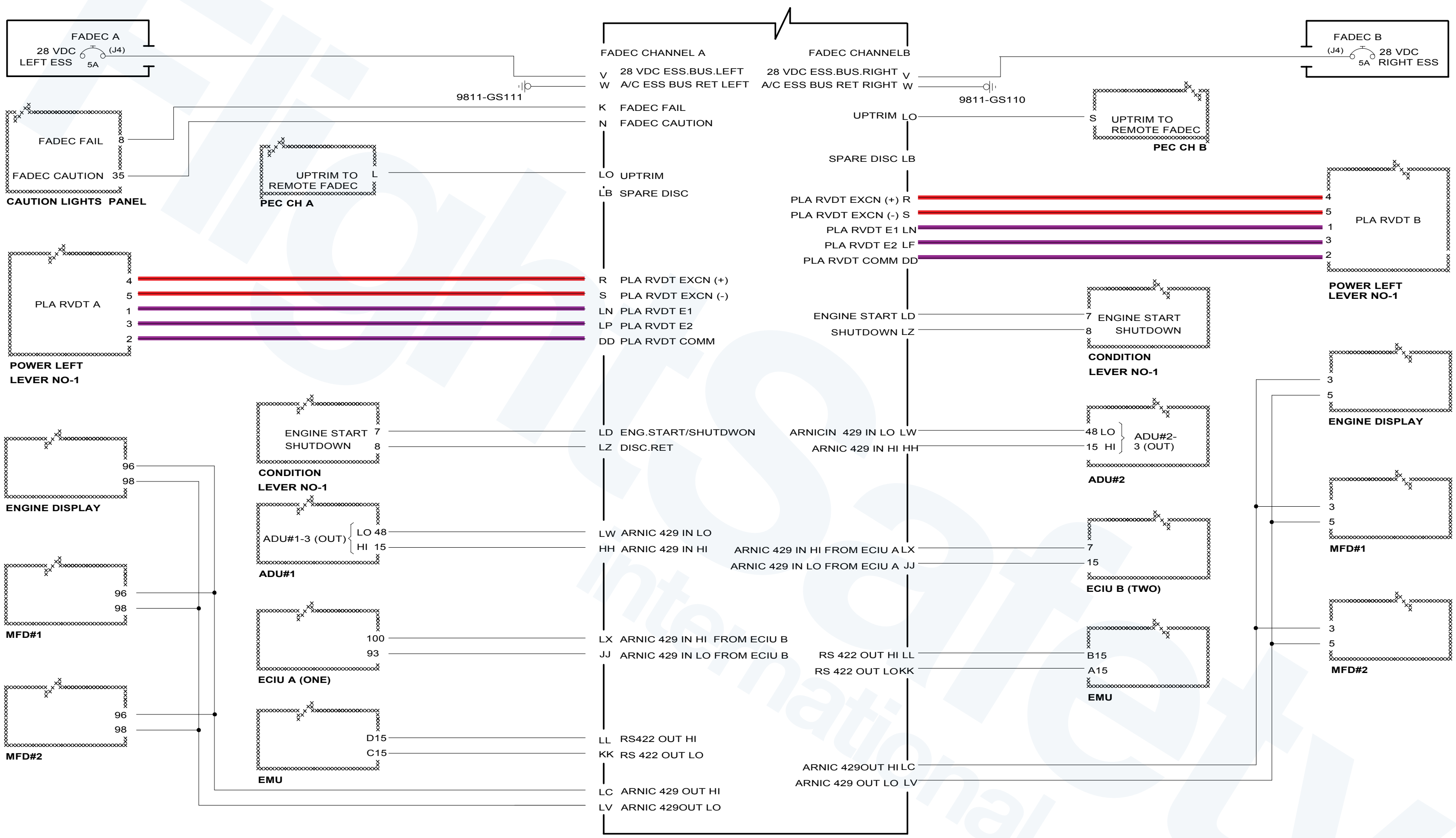


Figure 73-30. Engine Control Electrical Schematic

Throughout Engine Operation

Refer to Figure 73-30. Engine Control Electrical Schematic.

The PLA RVDTs receive their excitation power from FADEC and transmit their position back to FADEC.

Engine Start

PLA/DISC

When START selected:

- Ignition control signal sent from the ECIU to FADEC (see ignition operation)
- At approximately 20% N_H FADEC EEC power supply is transferred from the aircraft Essential bus to Permanent Magnet Alternator power
- FADEC monitors ITT, by signals from the MOT sensor
- CLA advanced to FEATHER/START (>15 degrees signal)
- If ITT is rising too high, too fast, FADEC EEC will decrease fuel flow to control ITT
- SHUTDOWN signal at both FADEC channels is removed
- Dual overspeed and shutdown solenoid in the FMU is de-energized
- P 2.7 HBOV closes
- FADEC commands dual torque motor of FMU to produce G.I. N_H 64.2%
- Feedback of metering valve position (FMU) is by LVDT

When CLA advanced to MIN, 900 or Max:

- CLA RVDT sends position signal to PEC
- PEC sends CLA position to FADEC by RS 422 data bus
- FADEC EEC indicates MCR, MCL, or NTOP on the Engine Display
- FADEC EEC sets torque gauge bugs to match power request.

When the PLA is advanced to Flight Idle:

- The FADEC EEC by controlling the FMU will increase the fuel flow to establish and maintain 660N_P as propeller blade angle increases from 0° to 16.5°.

When PLA advanced to RATING detent:

- Changing RVDT signals from PLA are received by FADEC EEC
- N_H, N_L, N_{PT} and Tq changing values are received by FADEC
- T 1.8 is received by local FADEC, then transmitted to the remote FADEC, the signals are averaged by both FADECs to set MTOP or NTOP
- Actual torque is matched to target torque (torque arrows at torque bug position)
- NTOP or MTOP is displayed in the top left and right corners of the ED
- The P2.2 HBOV is modulated in a closing direction from approximately 70% N_L and above.

If RDC TOP is selected, both channels of both FADECs must confirm, through cross communication, that the RDC TOP signal has been received. RDC TOP messages are displayed by FADEC EEC in the top corners of the ED. The torque bugs are repositioned by FADEC EEC to match the amount of reduction required.

After take-off at MTOP or NTOP, and CLA retarded to 900 position FADEC will:

- Recalculate and display required torque by rim bug position and digital readout on the torque gauges
- Control torque motor to control fuel to meet requirements
- Display MCL in top corners of the ED
- CLA retarded to MIN position
- FADEC controls engine to produce MCR as above.

Engine Failure during Takeoff at NTOP or RDC TOP.

There will be UPTRIM signal from the remote PEC.

The FADEC EEC will do the following:

- Increase torque bug position by 10% of selected power.
- Match actual torque to new torque bug position.
- Display UPTRIM message in top center of the ED.
- Display MTOP message in the appropriate top corner of the ED.
- If the bleed air is selected ON, and bleed flow selected to NORMAL or MAX, FADEC EEC will command MCP not MTOP.

Engine Failure During Take-Off at MTOP

With an UPTRIM signal from the remote PEC, the FADEC EEC will do the following:

- Display the UPTRIM message on the ED.

Alternate Power Settings

CLA/900
PLA/RATING

The above configuration would cause FADEC to produce MCL.

This can be changed by the operator to produce MCR at 900 Np.

This change is achieved by selecting the MCR discrete on the Engine Control Panel.

When MCR is selected at 900 N_P, FADEC will do the following:

- Display MCR in the top corners of the ED.
- Position the torque bug to the new calculated torque value.
- Control fuel from the FMU to produce the new torque.

CLA/850
PLA/RATING

The above configuration would cause FADEC to produce MCR. This can be changed by the operator to produce MCL at 850 N_P.

This change is achieved by selecting the MCL discrete on the Engine Control Panel.

When MCL is selected at 850 N_P, FADEC will do the following:

- Display MCL in the top corners of the ED
- Position the torque bug to the new calculated torque value
- Control fuel from the FMU to produce the new torque.

FADEC Fail

If FADEC fails the EEC will command its own FADEC FAIL warning light ON.

The following actions will occur:

- A signal output from FADEC EEC to the FADEC FAIL warning light
- The dual torque motor in the FMU is biased to close the flapper valve
- All PM pressure is removed from the full area of the metering valve
- The PR pressure acting on the half area will close the metering valve
- N_H will roll back to F.I, G.I. or Shutdown
- The operator will shut the engine down. CLA/FUEL OFF.

No.1 or No.2 ENG FADEC FAIL (Warning light)can be caused by the following:

- The two fuel flow feedback channels have malfunctioned
- The two ambient pressure channels have malfunctioned
- The two T1.8 temperature channels have malfunctioned
- The two N_H channels have malfunctioned
- A fuel flow wraparound and track malfunction occurs on a channel in control
- A malfunction is sensed during normal engine shutdown
- Metering valve malfunctioned during shutdown
- An N_L de-couple event has been sensed and accommodated
- The output driver test is enabled
- A critical fault indication by remote channel while crosslink is healthy.

FADEC Caution

If FADEC loses inputs to both channels, it will bring ON the FADEC caution light.

The signal to the caution light is from FADEC EEC Pin N.

The operator must do the following:

- Move the PLA of the affected engine slowly to avoid compressor stall/surge
- Power lever positions may need to be split to get the same power on both engines.

POWERPLANT Message

If the FADEC EEC detects a fault, or a number of faults, that would cause Loss Of Thrust Control (LOTC) it will bring on the POWERPLANT message. This message means NO DISPATCH.

It is important for the mechanic that the incorrect performance of some tests e.g. Prop Overspeed Test can bring on the POWERPLANT message.

Correct test performance will remove the message.

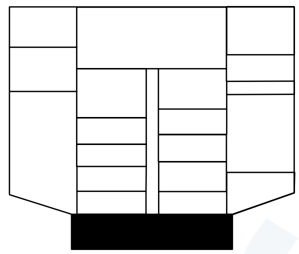
Fault Display

FADEC will display Powerplant and PEC faults when in maintenance mode that is selected by the MAINT DISC switch on the maintenance panel.

- FADEC channel A and PEC channel A faults will be displayed on the Torque Gauge
- FADEC channel B and PEC channel B faults will be displayed on the N_H gauge
- To move to the next fault code press MCL discrete on the engine control panel.
- Repeat until first recorded fault code returns.

REMEMBER other three codes will appear on the N_P and N_L gauges.

These are NOT fault codes, they are software codes. When you have retrieved all the fault codes, refer to the Fault Isolation Manual to find out what to do to fix the faults.



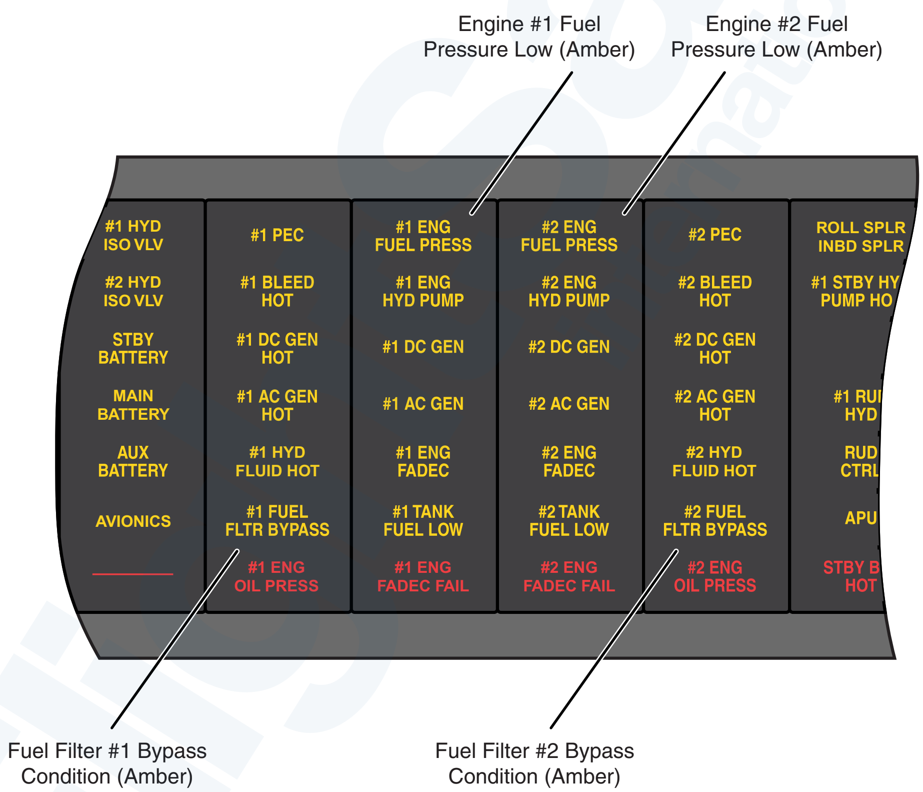


Figure 73-31. Caution and Warning Panel (CAWP)

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73-31-00 ENGINE FUEL INDICATION

GENERAL

Refer to Figure 73-31. Caution and Warning Panel (CAWP).

The fuel indicating system gets signals from switches to indicate Fuel System status.

The signals from the switches are processed in the FADEC.

The processed signals are sent to the Engine Cockpit Interface Unit (ECIU).

SYSTEM DESCRIPTION

The ECIU provides the following discrete outputs to the caution lights (on the Caution and Warning panel):

- No.1 ENG FUEL PRESS (Engine 1 Low Fuel Pressure)
- No.2 ENG FUEL PRESS (Engine 2 Low Fuel Pressure)
- No.1 FUEL FLTR BYPASS (Engine 1 Fuel Filter Impending Bypass)
- No.2 FUEL FLTR BYPASS (Engine 2 Fuel Filter Impending Bypass).

The fuel indicating system has the following switches:

- Low fuel pressure and
- Fuel filter impending bypass.

NOTES

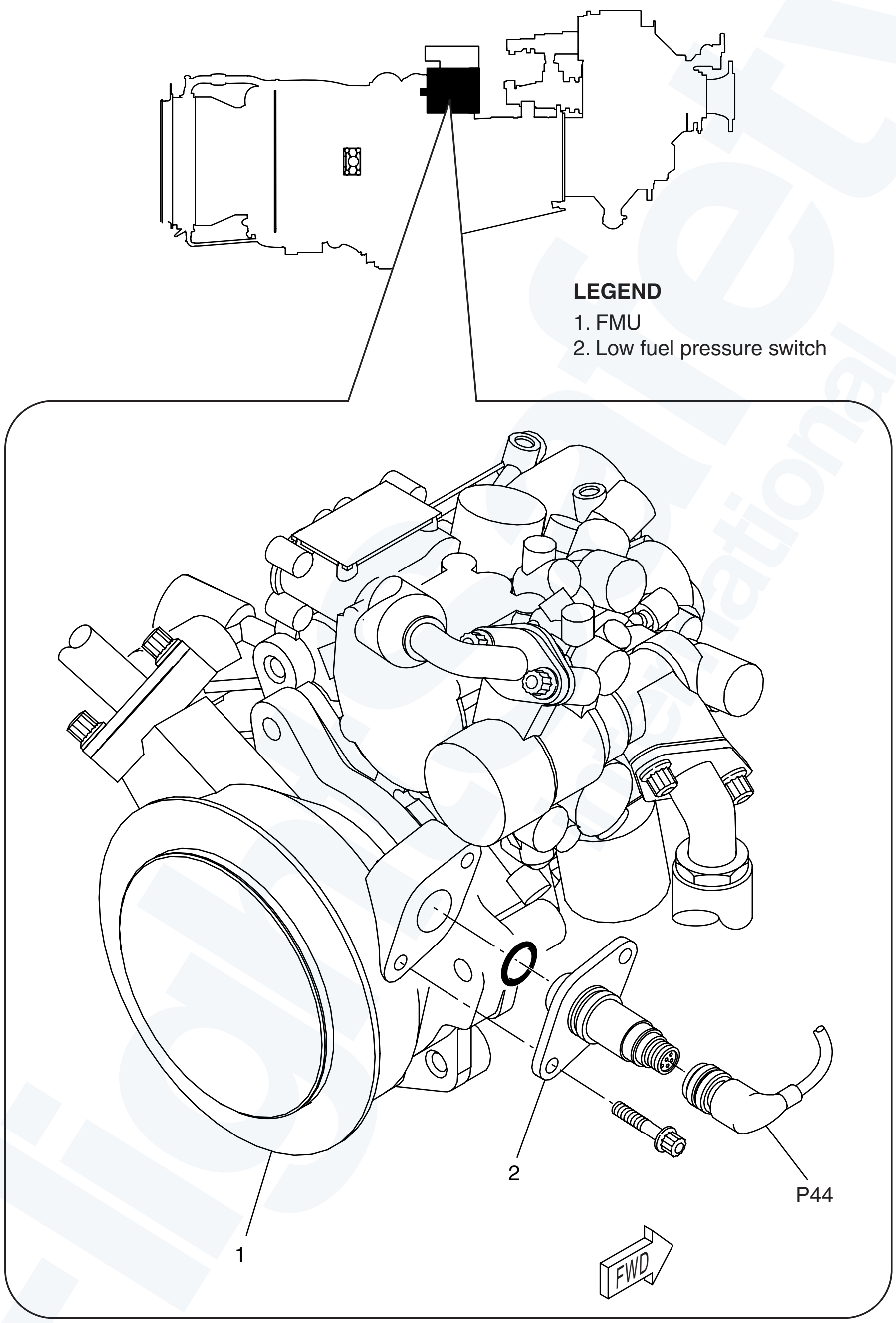

Figure 73-32. Low Fuel Pressure Switch

COMPONENT DESCRIPTION

Low Fuel Pressure Switch

Refer to Figure 73-32. Low Fuel Pressure Switch.

The low fuel pressure switch, installed on the FMU, indicates low fuel pressure at the inlet to the fuel pump.

Refer to Figure 73-33. Low Fuel Pressure Indication Block Diagram.

When the fuel pressure drops below the preset value, the electrical switch contacts relax to a closed state (resistance < 200 m Ω). This sends a signal to the FADEC, then to the ECIU and flight compartment for indication.

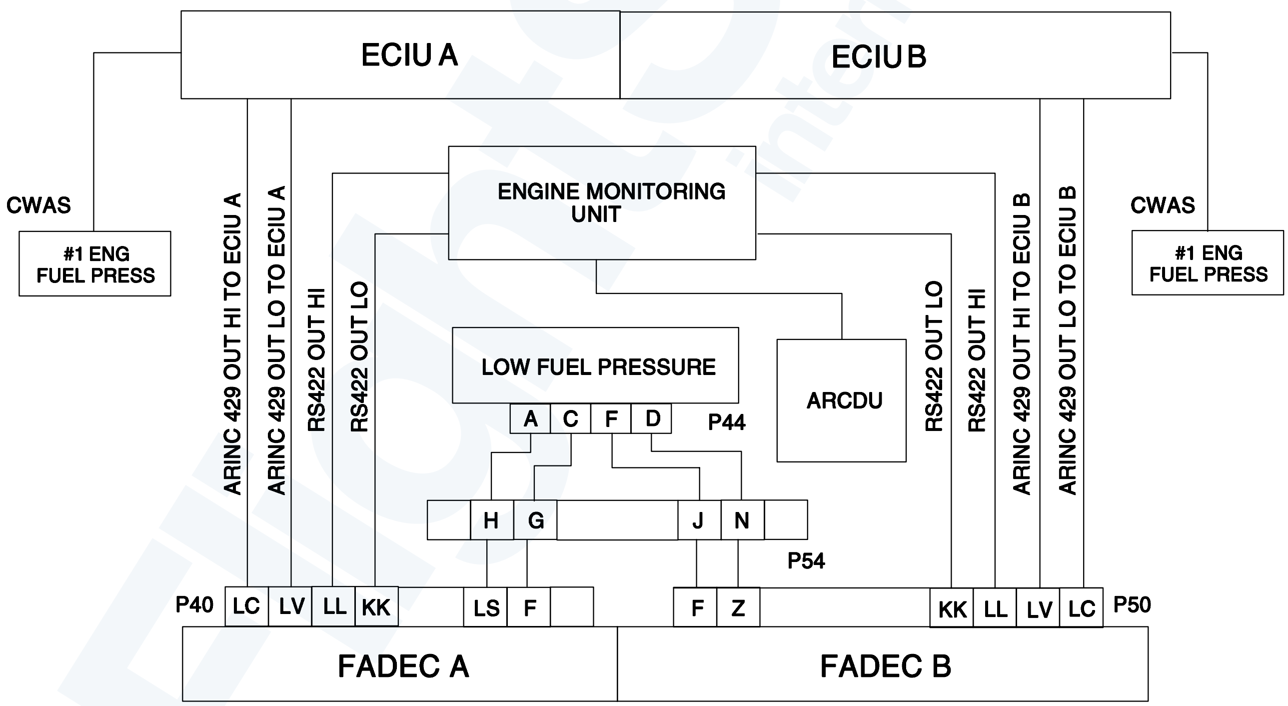

Figure 73-33. Low Fuel Pressure Indication Block Diagram

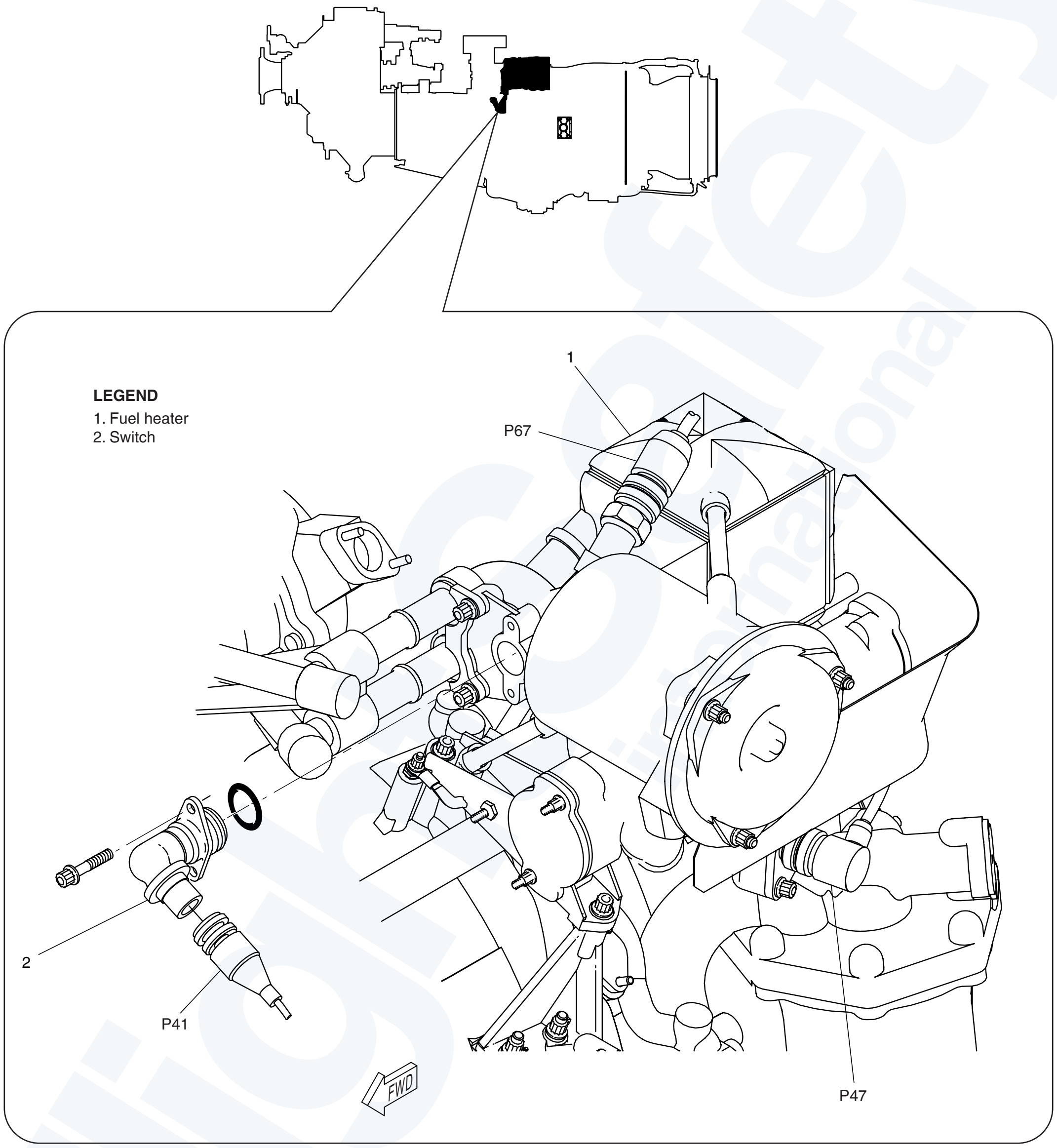

Figure 73-34. Fuel Filter Impending Bypass Switch

Fuel Filter Impending Bypass Switch

Refer to:

- Figure 73-34. Fuel Filter Impending Bypass Switch.
- Figure 73-35. Fuel Filter Bypass Indication Block Diagram.

The fuel filter impending bypass switch is installed on the fuel heater. It sends an indication of an impending bypass of unfiltered fuel.

If the differential pressure across the filter rises above a preset value, the electrical switch contacts are moved to the open state (switch resistance > 10 k).

This sends a signal to the FADEC, then to the ECIU and flight compartment for indication.

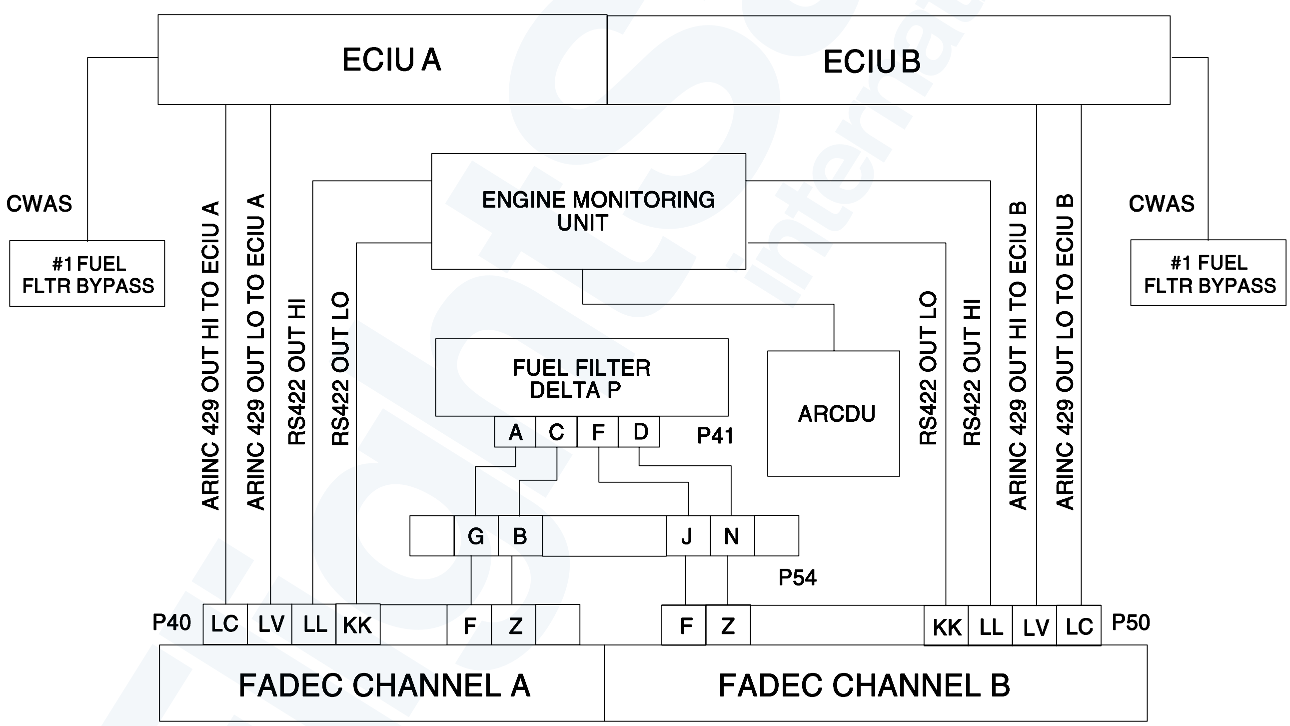

Figure 73-35. Fuel Filter Bypass Indication Block Diagram

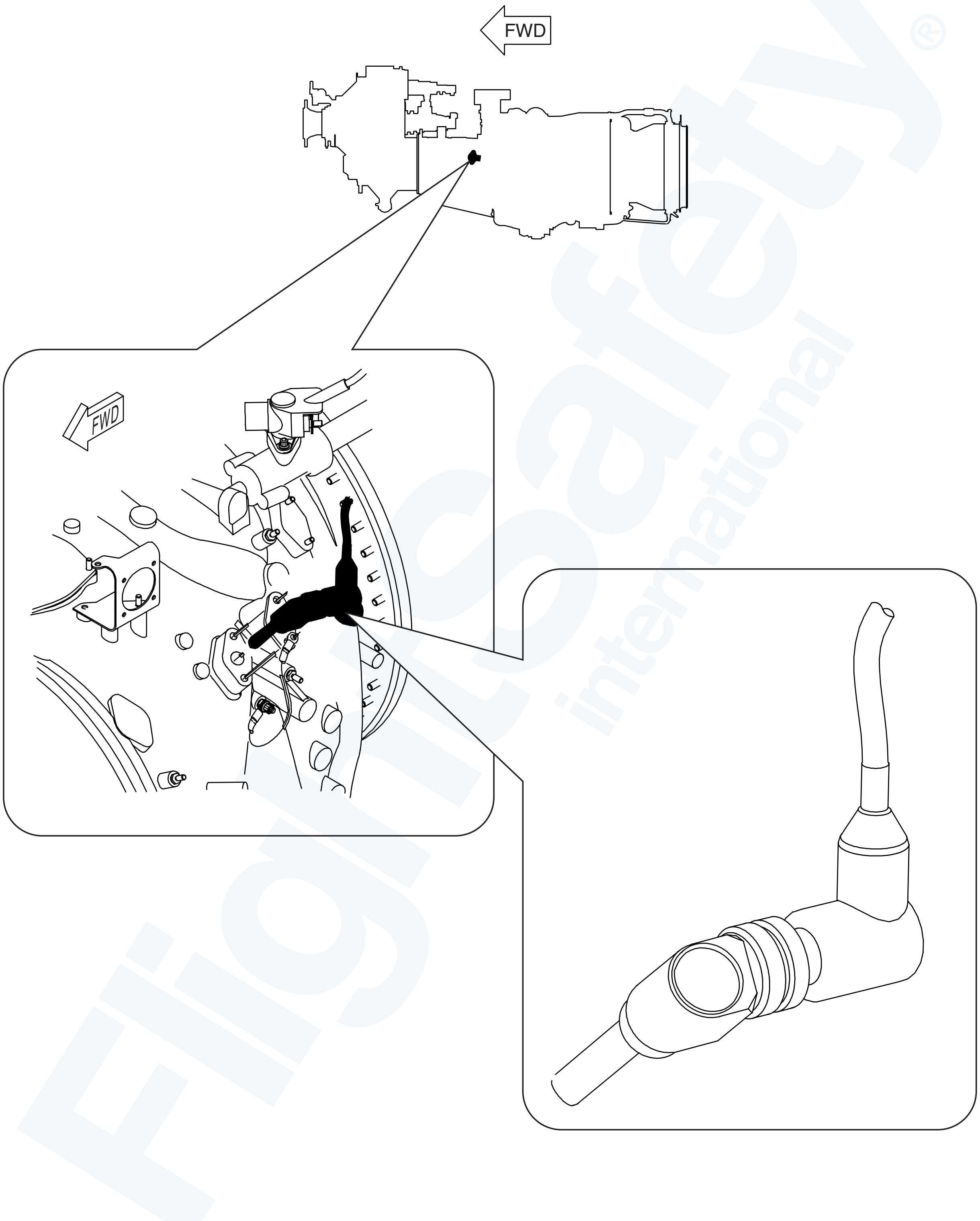

Figure 73-36. T1.8 Temperature Sensor

T1.8 Temperature Sensor

Refer to Figure 73-36. T1.8 Temperature Sensor.

A dual platinum resistance temperature sensor measures the intake air temperature.

The sensor is in the intake just upstream of the first stage low pressure compressor.

The signal from the sensor is proportional to the temperature.

It is used in calculating engine ratings to automatically set rated power and display the rated torque on the ED.

The signal is transmitted to the opposite FADEC and averaged by the two FADEC's

If the temperature difference between the 2 engines is less than 2 degrees C the FADEC's will compensate to ensure that RATED TORQUE is the same for both engines.

T1.8 is the primary source of intake temperature to FADEC for MTOP, NTOP and MCP. For the other selections the T1.8 is the backup for the ADC's.

FUNCTIONAL TEST OF THE T1.8 SENSOR

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

- De-energize the electrical system (TASK 24-00-00-861-802)
- Open the left side forward nacelle door
- Disconnect the control harness P14 connector from the sensor
- Visually inspect the sensor
- Using a digital ohmmeter do a continuity check
- Using a megohmmeter perform an insulation check
- If the values are beyond the limits for the tests replace the sensor
- Close the nacelle side door.

73-00-00 APPENDIX

MAINTENANCE CONSIDERATION

CDL

Refer to Configuration Deviation List (CDL) for the Q400 aircraft. These pages are contained in this appendix.

Unscheduled Inspection

Refer to the Bombardier published AMM Part 2 PSM 1-84-2.

- TASK 05-53-00-210-810 Engine Inspection after Chip Detector Circuit Completed
- TASK 05-53-00-750-802 Engine Inspection after Hydraulic Fluid in the Fluid System
- TASK 05-53-00-750-803 Fuel Nozzle Inspection after Use of Restricted Fuel

FADEC Fault Codes

Class 1 fault codes are associated with a crew advisory. For example a POWERPLANT message or ENG OIL PRESS warning light. This type of fault means No Dispatch of the aircraft is permitted until the fault(s) is addressed.

Class 2 fault codes are faults that do not produce a crew advisory and are considered as faults that are not Class 1 or Time Limited Dispatch faults.

There are three types of Class 2 fault codes as follows:

- Maintenance Fault Codes
- Information Fault Codes
- All other Class 2 Fault Codes.

Maintenance Fault Codes are associated with engine health sensors, chip detectors, filter bypass indicators and igniter plugs and maintenance action is recommended as soon as possible after the indication.

Information fault codes are associated with specific conditions noted during engine operation.

For example, compressor stall detected, slow engine start and PLA out of trim range. The requirement for maintenance action depends on the condition that was detected.

All other Class 2 faults codes mean maintenance action should be carried out as soon as it is practical.

Time Limited Dispatch Codes

NOTE

The dispatch time limits contained in the appropriate section of the P&W Airworthiness Manual cannot be increased and are not negotiable.

General

- The engine Time Limited Dispatch (TLD) allows the rectification of certain engine electronic control system faults to be delayed or deferred
- TLD generally applies to faults associated with control system functions which are redundant and will result in Loss of Thrust Control (LOTC) if the function is completely lost
- LOTC is defined as the complete or partial loss of control over engine thrust (not necessarily an in-flight shutdown)
- The TLD risk, or weighting number represents a margin of safety allowed for some engine faults.

Levels of TLD

There are four levels of TLD. These are:

- Fully operational system - no faults, no dispatch restrictions
- Long Term Dispatch (LTD) - Dispatch is permitted up to 500 FH after the fault occurs
- Short Term Dispatch (STD) - Dispatch is permitted up to 150 FH after the fault occurs
- No Dispatch (ND) - No dispatch is permitted with an existing combination of faults.

Fully Operational System

A system is considered fully operational when no faults are annunciated.

Long Term Dispatch

The control system is on a long term dispatch if the sum of the TLD risk associated with the annunciated faults is between 10 and 74.9.

Short Term Dispatch

The control system is on a short term dispatch, if the sum of the TLD risk associated with the annunciated faults is equal to 75 or more.

The TLD risk, or weighting numbers, are programmed into the Engine Monitoring Unit and can also be found in Pratt and Whitney Airworthiness Manual if TLD is being calculated manually.

No Dispatch

The FADEC calculates the resistance of the powerplant to Loss Of Thrust Control (LOTC). When it determines that there is a danger of LOTC it will cause the POWERPLANT message to come into view on the Engine Display. This message means that no more flights are allowed until at least some of the TLD faults present are corrected.

APPROVED FUELS FOR THE PW150 ENGINE

All approved fuels for pre and post SB 35189 engines can be found in this TASK.

73-00-00 SPECIAL TOOLS & TEST EQUIPMENT

- PWC57154 Fork
- Commercially available Ultrasonic or electrosonic cleaning equipment
- PWC32396 Screws, Jacking
- PWC30128 - 05 Puller
- Barfield TT-1000A Ohmmeter/Insulation Tester
- Simpson (or equivalent) Ohmmeter
- Commercially Available Ohmmeter, Digital
- PWC42034 Jackscrew
- PWC55143 Base
- PWC55147 Socket
- PWC57084 Holder
- PWC58104 Wrench, mini-strap Glenair TG69 Pliers, soft-jawed

73-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 73-11-31-000-801: Removal of the Fuel Filter.
- AMM 73-11-31-400-801: Installation of the Fuel Filter.
- AMM 73-11-31-610-801: Servicing After Fuel Filter Bypass Extension.
- AMM 73-21-01-000-801: Removal of the FADEC.
- AMM 73-21-01-400-801: Installation of the FADEC.
- AMM 73-21-06-000-801: Removal of the Fuel Metering Unit.
- AMM 73-21-06-400-801: Installation of the Fuel Metering Unit.
- AMM 71-56-01-000-801: Removal of the FADEC Electrical Wiring Harness (Harness 82454109).
- AMM 71-56-01-400-801: Installation of the FADEC Electrical Wiring Harness (Harness 82454109).
- AMM 73-21-11-000-801: Removal of the Fuel Control Electrical Wiring Harness (PWC Harness 3122401).
- AMM 73-21-11-400-801: Installation of the Fuel Control Electrical Wiring Harness (PWC Harness 3122401).
- AMM 73-21-16-000-801: Removal of the Characterization Plug.
- AMM 73-21-16-400-801: Installation of the Characterization Plug.
- AMM 73-21-21-000-801: Removal of the Permanent Magnet Alternator.

- AMM 73-21-21-400-801: Installation of the Permanent Magnet Alternator.
- AMM 45-00-73-742-801: Retrieval of Data from the Central Diagnostic System (CDS) - Engine Monitoring (EMU).
- AMM 73-21-06-720-801: Functional Test of the Fuel Metering Unit
- AMM 73-21-00-070-801: FADEC Fault Code Clear.
- AMM 73-21-00-070-802: Fault Code Clear Following FADEC Replacement.
- AMM 73-21-00-740-801: FADEC ID Check.